



MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY
MALAWI INSTITUTE OF TECHNOLOGY

Computer Science & Information Technology Department (CSIT)

MID - SEMESTER EXAM

COMP-122: INTRODUCTION TO COMPUTER PROGRAMMING

DATE: 2nd December 2020

Time: 9:00 – 12:00 [3 hours]

INSTRUCTIONS

1. This paper has FIVE (5) pages, please check.
2. Answer all questions.
3. Marks for each question are indicated on the right margin.
4. Write your registration number on top of each answer sheet provided.
5. Phones and PCs of any kind are not allowed.

7, 8, 9, 10

1. What do you understand by the following terms? [6]
- (a) Machine language
 - (b) Assembly language
 - (c) High level language
2. What is a compiler? Why is it needed to run a program written in C++? [4]
3. Declare a variable for each of the following quantities: [8]
- (a) A person's body temperature in degrees centigrade.
 - (b) Volume of a bucket.
 - (c) Number of machines in a workshop.
 - (d) For writing values like 'A', 'B' etc. in a program.
4. Evaluate the following expression. Show all your steps. [5]
- $(1 > 0) \parallel (1 - 2 > 0) \&\& (3 == 5)$
5. Fix all errors (by rewriting the program) in the following program either logical or syntactical or that might cause runtime errors: [10]

```
int Main(void)
    int a = 2;
    int b = 5;
    int c == a;    / assign c to a
    return;
}
```

6. Determine the output of following program.

[12]

```
#include <iostream>

using namespace std;

int main ()
{
    int m=3, n=4, p=5, s=6, q=31, k=10;
    m += 2;
    n *= 4;
    p -= 3;
    s /= 2;
    k *= ++k;
    cout << "k = " << k << ", " << "m = " << m << ", n = " <<
    n << ", p = " << p << ", s = " << s << endl;
    q %= 4 + (n /= 4);
    cout << " q = " << q << endl;
    return 0;
}
```

7. Write a program to read 10 integers and to display their average value. Use a "for loop" to read all the integers and calculate the average after the 10th integer is captured. [11]

8. Write a program to prove numerically that $(A + B)^2 = A^2 + B^2 + 2 A B$. [11]

9. Write a Boolean *if (condition)* for the following situations.

(a) B varies between 20 to 30 and between 40 to 50.

[4]

(b) A varies from 0 to 10 but not between 4 to 6.

[4]

10. Write a program that asks a user to enter three numbers then finds and displays the biggest number. [11]

11. Study the program below, and rewrite it using the "switch-case" structure. [9]

```
#include<iostream>
using namespace std;

int main()
{
    int m;
    cout << "Name the day which is day number ";
    cin >> m;
    if (m == 1) cout<< "Monday";
    else if (m == 2) cout << "Tuesday";
    else if (m == 3) cout << "Wednesday";
    else if (m == 4) cout << "Thursday";
    else if (m == 5) cout << "Friday";
    else if (m == 6) cout << "Saturday";
    else if (m == 7) cout << "Sunday";
    else cout << "The number entered is not in the range.";
    return 0;
}
```

12. Rewrite the program below using the "*while loop*".

[6]

```
#include <iostream>

using namespace std;

int main()
{
    int i, sum=0;
    for ( i=0; i<=10; i++;)
    {
        sum += i;
    }
    cout << "sum = " << sum << endl;
    return 0;
}
```



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DEPARTMENT OF COMPUTER SCIENCE AND INFORMATION TECHNOLOGY

END OF SEMESTER EXAM

COMP-122: INTRODUCTION TO COMPUTER PROGRAMMING

DATE: 18th December 2020

Time Allowed: 3 hours

EXAMINERS: R. Hiwa, R. Tambala, A. Chongwe, W. Gondwe, P. Katengeza, L. Msosa

INTERNAL MODERATOR: Dr. B. Kankuzi

INSTRUCTIONS

1. This paper has 6 pages, please check.
2. Answer ALL questions.
3. Marks for each question are indicated on the right margin. The total is 100 marks.
4. Write your registration number on top of each answer sheet provided.
5. Reference materials, phones, PCs or devices of any kind are not allowed.

SECTION A: Multiple choice

I. What is the output after executing the following segment of code?

```
int num = 10;  
if (num > 10)  
    cout << "Yes";  
else  
    cout << "No";
```

A. Yes

C. No

B. YesNo

D. Nothing

II. A single-alternative selection is one in which an action takes place _____.

A. only when the result of the decision is true

B. only when the result of the decision is false

C. when the result of the decision is either true or false

D. when the result of the decision is both true and false

III. Which of the following does not have the same result as:

```
myValue = myValue + 1
```

A. myValue = 1;

B. myValue += 1;

C. ++myValue;

D. myValue++;

IV. In C++, all false relational expressions are evaluated as _____.

A. 0

B. Negative

C. 1

D. positive

V. If c and d are integers, and $d = 34$, what is the value of $c = d++$?

- A. 0 B. 1 C. 34 D. 35

VI. If a and b are integers, and $a = 10$ and $b = 30$, if you use the statement $a += b$, what is the resulting value of a ?

- A. 10 B. 20 C. 30 D. 40

VII. In C++, what is the result of $19 \% 5$?

- A. 3 B. 3.8 C. 4 D. Error

VIII. You use a _____ to create a compound Boolean expression in which two conditions must be true for the entire expression to evaluate as true.

- A. logical AND operator C. switch statement
B. logical OR operator D. any of the above

IX. In a while loop, _____.

- A. a Boolean expression is evaluated; if it is true, the loop is over
B. the loop body always consists of a single statement
C. the loop body always executes at least once
D. the Boolean expression is always evaluated at least once

X. A loop control variable should almost always be used *three* times in these ways:

- A. initialize, display, compare
B. initialize, compare, alter
C. display, alter, stop
D. loop, alter, display

XI. Which one of the following is **NOT** a fundamental data type in C++?

- A. float B. int C. string D. bool

SECTION B: Structured questions

2. List any *three* computer programming languages you know except C++. [3]
3. Below are two declarations in the header section of a C++ program. Explain what each of the statements is doing.
 - a) `#include <iostream>` [2]
 - b) `using namespace std;` [2]

4. Write a program that uses a **for loop** to print a multiplication table of 7 from 1 to 12. The output should look like the following:

1 x 7 = 7

2 x 7 = 14

...

12 x 7 = 84

Handwritten notes and calculations:

```

int i, int j; a = 7;
for (i = 1; i <= 12; i++)
{
    for (j = 1; j <= 7; j++)
    {
        cout << i * j << " ";
    }
    cout << endl;
}
  
```

Other calculations shown:

- $1 \times 7 = 7$
- $2 \times 7 = 14$
- $1 \times 6 = 6$
- $2 \times 6 = 12$
- $1 \times 7 = 7$
- $2 \times 7 = 14$
- $1 \times 6 = 6$
- $2 \times 6 = 12$
- $1 \times 7 = 7$
- $2 \times 7 = 14$
- $1 \times 6 = 6$
- $2 \times 6 = 12$
- $1 \times 7 = 7$
- $2 \times 7 = 14$
- $1 \times 6 = 6$
- $2 \times 6 = 12$

5. Given the following code:

```

int x;
cin >> x;
if (condition) {
    cout << "Yes!";
}
  
```

Handwritten notes:

- $x == +$ // $x == 0$
- $x \% 2 = 0$

Write proper code to achieve the following conditions:

- a) check if x is positive or zero [2]
- b) check if x is an even number [2]
- c) check if x subtracted by three is not equal to nine [2]
- d) check if x is a multiple of four and five [5]

6. Rewrite the following program to correct all syntax errors:

[10]

```
#include <iostream>

main {
    cout << 'Zatheka!';
    RETURN 0;
}
```

7. Given the following code:

```
if ( entry == 'F') {
    cout << "Female";
} else if ( entry == 'M') {
    cout << "Male";
} else {
    cout << "Not set";
}
```

Write an equivalent code using the switch statement.

[10]

8. Mention any *two* benefits of computer programming skills.

[4]

9. Give *one* advantage that machine language has over high-level language. [2]

10. Give any *one* advantage that high-level language has over machine language. [2]

11. Write a statement for:

a) Declaring an integer variable called **number**.

[2]

b) Assigning a boolean variable **flag** to **true**.

[2]

c) Initialising a character variable called **vowel** to 'a'.

[3]

12. Consider the following code:

```
for (int x = 100; x >= 0; x -= 10) {
    cout << x;
}
```

Write an equivalent code using a do while loop.

[6]

13. Write a program to read n integers and to display their average value. Use a "for loop" to read all the integers and calculate the average after the nth integer is captured. The program should allow the user to enter their chosen n (integer), it should also ask and allow user to enter n-integers, and finally it should print the computed average of n numbers.

[10]

14. Write a program to compare three numbers and find the smallest among them.

[10]

THIRTEEN

for

int n; x n-w x >=

for(a = a >= 20, 2 3 4

2 >= 3 || 2 <= 3

END OF QUESTION PAPER

int n;

if n for (n = 0, n < 10, n++) if (num1 >

{ cout << n << endl; num1 = 5

if (num1 >= num2)

Reg. No.: _____



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MALAWI INSTITUTE OF TECHNOLOGY
COMPUTER SCIENCE DEPARTMENT
2018/19 ACADEMIC YEAR
END OF SEMESTER EXAMINATION
COMP-122: COMPUTER PROGRAMMING

DATE: 30th JULY, 2019

Time: 09:00 – 12:00

Examiners: C. Mtegha-Mkohongo, K. Gondwe, R. Hiwa, A. Chongwe, K. Makanda, A. Mwangonde, M. Phiri

INSTRUCTIONS

- A. This paper has 16 pages, please check.
- B. Answer all questions.
- C. Marks for each question are indicated on the right margin.
- D. Write your registration number on top of each answer sheet provided.
- E. Start each question on a fresh page and write neatly.

Reg. No.: _____

QUESTION ONE

[20]

- a. State four rules for naming variables.

[4]

- b. Find the correct identifiers out of the following, which can be used for naming variable, constants or functions in a C++ program:
- While, for, Float, break, 2ndName, A%B, Amount2, _Counter

[4]

- c. What using statement do you always put at the top of your programs (to get standard naming conventions)?

[2]

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- d. Write a Pseudocode to add two numbers that are entered by the user and shows the result. [6]

- e. Briefly describe the difference between a function declaration (prototype) and a function definition. [4]

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[30]

QUESTION TWO

a. What is the output of following code fragments?

[1]

```
i.  int result = 4 + 5 * 6 + 2;
    cout << result;
```


[1]

```
ii. int a = 5 + 7 % 2;
    cout << a;
```


[2]

```
iii. int x = 10, y;
      y = x++;
      cout << y;
```


[2]

```
iv.  int x = 10, y;
      y = x++;
      cout << x;
```


[2]

```
v.   int x = 10;
      y = ++x + x++ + x;
      cout << y;
```


Reg. No.: _____

```
x. int n;
   for (n = 5; n > 0; n--)
   {
       cout << n;
       if (n == 3){
           break;
       }
   }
```

```
xi.  int i = 5;                                     [2]
      int j = i++;
      cout<<j<<i;
```

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- b. Given the following program: if a user runs the program and input 27/7/17, what would the output be? [3]

```
#include<iostream>
using namespace std;
int main (){
    int day, month, year;
    char ch;
    cout << "Input a date as dd/mm/yy: ";
    cin >> day >> ch >> month >> ch >> year;

    switch (month) {
        case 1:      cout << "January"; break;
        case 2:      cout << "February"; break;
        case 3:      cout << "March"; break;
        case 4:      cout << "April"; break;
        case 5:      cout << "May"; break;
        case 6:      cout << "June"; break;
        case 7:      cout << "July"; break;
        case 8:      cout << "August"; break;
        case 9:      cout << "September"; break;
        case 10:     cout << "October"; break;
        case 11:     cout << "November"; break;
        case 12:     cout << "December"; break;
    }
    cout << " " << day << ", " << 1900 + year << "\n";
    return 0;
}
```

Reg. No.: _____

c.

```
int choice = 1;
switch(choice)
{
    //cout << "\nFirst Statement";
    case 1:
        cout << "\nInside Switch case 1";
    case 2:
        cout << "\nInside Switch case 2";
        break;
    case 3:
        cout << "\nInside Switch case 3";
        break;
    default:
        cout << "bye bye";
}
```

[2]

```
d. int a = 0, b;
    b = (a = 50) + 10;
    cout << a << "+" << b;
```

[3]

Reg. No.: _____

QUESTION THREE

[40]

a. What does the following program do?

[4]

NOTE: Do NOT explain how it does it.

```
#include<iostream>
using namespace std;
int main() {
    int Arr[100], n, i, temp;
    cin >> n;
    for (i = 0; i < n; i++)
    {
        cout << "Enter element " << i + 1 << ": ";
        cin >> Arr[i];
    }
    temp = Arr[0];
    Arr[0] = Arr[n - 1];
    Arr[n - 1] = temp;
    for (i = 0; i < n; i++)
        cout << Arr[i] << " ";
    return 0;
}
```

BA

- 11

12

Reg. No.: _____

- c. Write a program using function called *addition* which accepts TWO integers as arguments and returns their sum. Call this function from `main( )` and print the results in `main( )`. [6]

11 | Page

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d. Write a program which accept two numbers and print their product.

[4]

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- e. Write a program to swap the values of two variables without using third variable [8]

13 | Page

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f. Write a C++ program to find the largest and smallest element of an array of size 100. [10]

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QUESTION FOUR

[10]

a. Given an array of y elements, what is the index of:

i. The first Element

[1]

ii. The Last Element

[1]

b. Write a C++ program that declares an array numbers of 100 components of type int. Initialize the array so that the first 50 components are equal to the square of the index variable, and the last 25 components are equal to three times the index variable. [5]

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- c. How many minimum number of functions should be present in a C++ program for its execution? [1]

- d. What is the scope of the variable declared in the user defined function? [1]

End of Question Paper

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DEPARTMENT OF APPLIED STUDIES
END OF SEMESTER EXAMINATION

CHEM-122: CHEMISTRY II

DATE: 6th AUGUST, 2018

TIME: 9:00AM-12:00Noon

INSTRUCTIONS

1. This examination paper has five questions.
2. Answer all questions.
3. This examination paper contains 4 pages including this cover page.
Please check.
4. Write your Registration Number on each answer sheet.
5. All working must be clearly shown.
6. Marks for each question are indicated against it.
7. The Periodic Table of the elements is provided.

The following gas constants may be useful:

$$R=8.314\text{J/mol.k}$$

$$R=0.0826 \text{ atm.L/mol.k}$$

08260



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DEPARTMENT OF APPLIED STUDIES
END OF SEMESTER EXAMINATION

CHEM-122: CHEMISTRY II

DATE: 6TH AUGUST, 2018

TIME: 9:00AM-12:00Noon

INSTRUCTIONS

1. This examination paper has five questions.
2. Answer all questions.
3. This examination paper contains 4 pages including this cover page.
Please check.
4. Write your Registration Number on each answer sheet.
5. All working must be clearly shown.
6. Marks for each question are indicated against it.
7. The Periodic Table of the elements is provided.

The following gas constants may be useful:

$$R=8.314\text{J/mol.k}$$

$$R=0.0826 \text{ atm.L/mol.k}$$

08260

specific heat cap iron = $0.45 \text{ J/g}^\circ\text{C}$
 specific heat cap of water = $4.184 \text{ J/g}^\circ\text{C}$
 q fusion = $0.903 \text{ J/g}^\circ\text{C}$

QUESTION ONE

- a) State Boyle's law? [2 Marks]
- b) The ideal gas equation $PV = nRT$ is a combination of different gas laws. Deduce the ideal gas equation from the three gas laws. [4 Marks]
- c) A balloon is filled with an ideal gas at a temperature of 27°C at 1 atm of pressure. If the balloon is heated to 127°C at constant pressure, by what factor does the volume change? [4 Marks]
- d) A 502.8 g sample of X_2 (g) has a volume of 9.0 L at 10 atm and 102°C . What element is X? [4 Marks]
- e) What is the density of a gas with molar mass of 100g/mol at 0.5 atm and 27°C ? [4 Marks]

$$P_1 V_1 = P_2 V_2$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2} \times \frac{T_2}{T_1}$$

$$\frac{P_1 T_2 V_1}{P_2 T_1 V_2}$$

QUESTION TWO

- a) If a 40.1 g piece of iron at 652°C is dropped into a sample of 328 g of water at 32.4°C , what will the final temperature be after thermal equilibrium is established? Assume that no heat is lost during the process. [6 Marks]
- b) A 32.5 g cube of aluminum initially at 45.8°C is submerged into 105.3 g of water at 15.4°C . What is the final temperature of both substances at thermal equilibrium? (Assume that the aluminum and the water are thermally isolated from everything else.) [4 Marks]

$$q = q = m \times C \times \Delta T$$

$$d = \frac{m}{V} = \frac{m}{m}$$

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H_2

$$PV = nRT$$

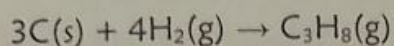
$$P_1 V_1 = nRT$$

$$d_{\text{mass}} = \frac{\text{mass}}{\text{volume}}$$

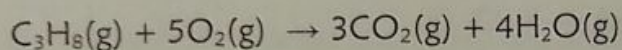
- c) When 1.515 g of glucose ($C_6H_{12}O_6$) undergoes combustion in a bomb calorimeter, the temperature rises from $23.12^\circ C$ to $29.33^\circ C$. Find E_{rxn} for the combustion of glucose in kJ/mol glucose. The heat capacity of the bomb calorimeter, determined in a separate experiment, is $4.90 \text{ kJ}/^\circ C$. [6 Marks]

- d) Find H_{rxn} for the reaction:

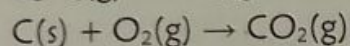
[6 Marks]



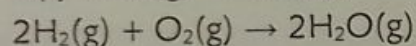
Use these reactions with known H 's:



$$H = -2043 \text{ kJ}$$



$$H = -393.5 \text{ kJ}$$

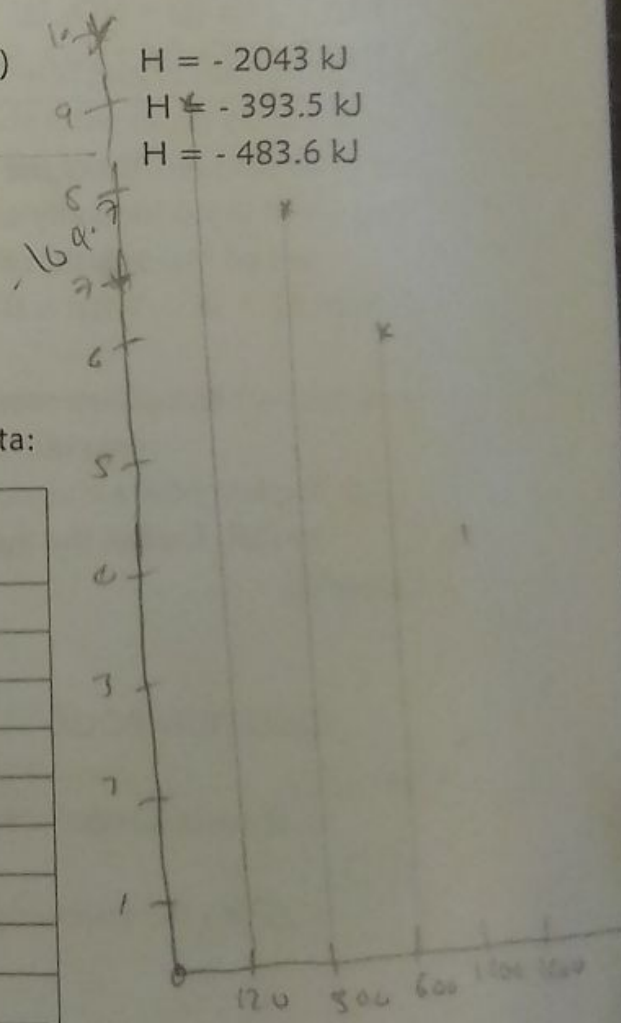


$$H = -483.6 \text{ kJ}$$

QUESTION THREE

- a) Based on the following experimental data:

Time (s)	$[H_2O_2]$ (mol/L)
0	1.0
120	0.91
300	0.78
600	0.59
1200	0.37
1800	0.22
2400	0.13
3000	0.082
3600	0.050



- i. What is the order of this reaction with respect to the reactant H_2O_2 , and how do you know? [8 Marks]
- ii. What is the form of the integrated rate law? [2 Marks]
- iii. What is the value of the rate constant (k)? [3 Marks]

b) The reaction mechanism for a reaction is as follows

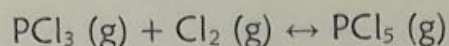
Step 1	$\text{O}_3 (\text{g}) + \text{NO}_2 (\text{g}) \rightarrow \text{NO}_3 (\text{g}) + \text{O}_2 (\text{g})$	slow
Step 2	$\text{NO}_3 (\text{g}) + \text{NO}_2 (\text{g}) \rightarrow \text{N}_2\text{O}_5 (\text{g})$	fast

- i. Write the overall reaction [2 Marks]
 - ii. Identify all that apply as: reactants, products, catalysts, and reaction intermediates [4 Marks]
 - iii. What is the rate law expression for this mechanism? [2 Marks]
 - iv. What would the concentration of H_2O_2 be at 4500 seconds? [2 Marks]
- c) Explain how a chemical reaction occurs according to the collision model. Explain the meaning of the orientation factor in this model. [4 Marks]

QUESTION FOUR

- a) State Le Châtelier's principle. [2 Marks]
- b) Given the following reversible reaction:

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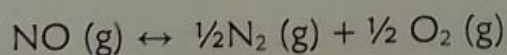
i) What will happen if some extra PCl_3 were added to the reaction container? [2 Marks]

ii) What will happen if the total pressure of the container is increased? [2 Marks]

iii) Suppose for the above reaction we keep the equilibrium concentrations of the reactants $[\text{PCl}_3] = 0.5\text{M}$ and $[\text{Cl}_2] = 0.07\text{M}$ and the temperature at 400K so that the $K_c = 96.2\text{M}^{-1}$. What is the equilibrium concentration of the product? [4 Marks]

c) Nitrogen monoxide is a major cause of acid rain in the atmosphere. One major source of nitrogen monoxide is the direct combination of nitrogen and oxygen in automobile engines. The value of K_c for this reaction is 1.7×10^{-3} at 2300K . If a mixture of 0.319M N_2 gas and 0.0859M O_2 gas is heated to 2300K , what will be the equilibrium concentration of NO ? [6 Marks]

d) Using the above information (c) compute the equilibrium constant value for the following reaction at 2300K .



[3 Marks]

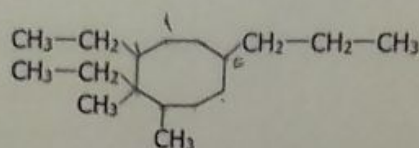
QUESTION FIVE

← =

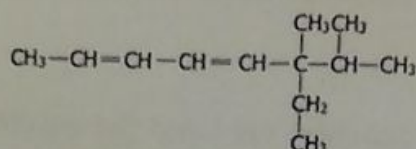
a) Draw two molecules that are structural isomers. [4 Marks]

b) Using the IUPAC system, name the following compounds: [3 Marks]

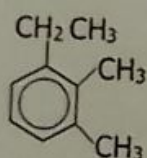
i.



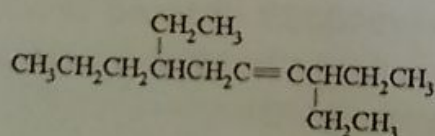
ii.



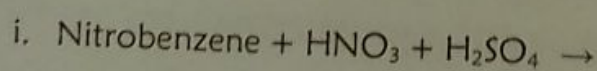
iii.



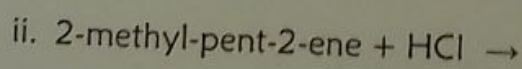
iv.



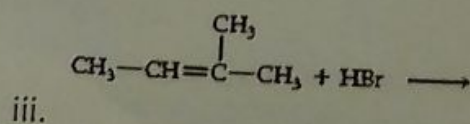
c) Write the products of the following organic reactions:



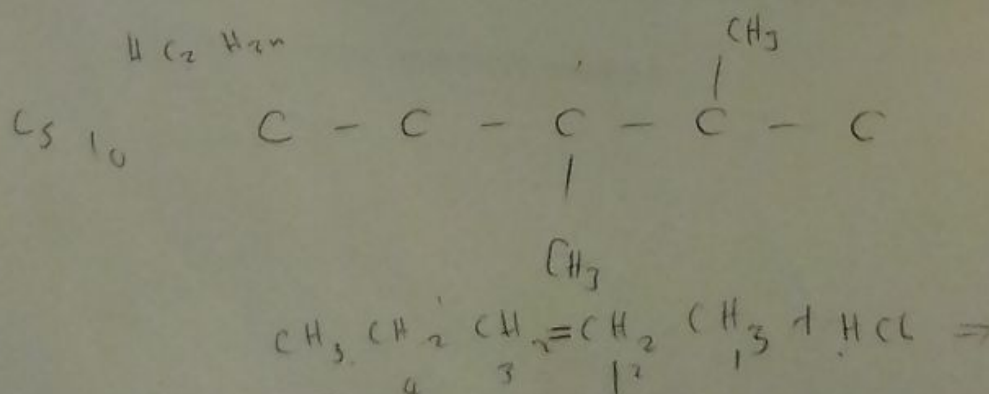
[2 marks]



[2 marks]



[2 Marks]



The Periodic Table of the Elements

element name	atomic number	symbol	atomic mass
hydrogen	1	H	1.008
lithium	3	Li	6.941
beryllium	4	Be	9.012
sodium	11	Na	22.99
magnesium	12	Mg	24.31
calcium	20	Ca	40.08
scandium	21	Sc	44.96
titanium	22	Ti	47.88
vanadium	23	V	50.94
chromium	24	Cr	52.00
manganese	25	Mn	54.94
iron	26	Fe	55.85
cobalt	27	Co	58.93
nickel	28	Ni	58.69
copper	29	Cu	63.55
zinc	30	Zn	65.39
gallium	31	Ga	69.72
germanium	32	Ge	72.59
arsenic	33	As	74.92
selenium	34	Se	78.96
bromine	35	Br	79.90
krypton	36	Kr	83.80
rubidium	37	Rb	85.47
strontium	38	Sr	87.62
yttrium	39	Y	88.91
zirconium	40	Zr	91.22
niobium	41	Nb	92.91
molybdenum	42	Mo	95.94
technetium	43	Tc	[98]
ruthenium	44	Ru	101.1
rhodium	45	Rh	102.9
palladium	46	Pd	106.4
silver	47	Ag	107.9
cadmium	48	Cd	112.4
indium	49	In	114.8
tin	50	Sn	118.7
antimony	51	Sb	121.8
tellurium	52	Te	127.6
iodine	53	I	126.9
xenon	54	Xe	131.3
cesium	55	Cs	132.9
barium	56	Ba	137.3
lanthanum	57	La	138.9
cerium	58	Ce	140.1
praseodymium	59	Pr	140.9
neodymium	60	Nd	144.2
promethium	61	Pm	[145]
samarium	62	Sm	150.4
europium	63	Eu	151.9
gadolinium	64	Gd	157.3
terbium	65	Tb	158.9
dysprosium	66	Dy	162.5
holmium	67	Hm	164.9
erbium	68	Er	167.3
thulium	69	Tl	168.9
ytterbium	70	Yb	173.0
lutetium	71	Lu	174.9
hafnium	72	Hf	178.5
tantalum	73	Ta	180.9
tungsten	74	W	183.9
rhenium	75	Re	186.2
osmium	76	Os	190.2
iridium	77	Ir	192.2
platinum	78	Pt	195.1
gold	79	Au	197.0
mercury	80	Hg	200.6
thallium	81	Tl	204.4
lead	82	Pb	207.2
bismuth	83	Bi	209.0
polonium	84	Po	[210]
astatine	85	At	[210]
radon	86	Rn	[222]
francium	87	Fr	[223]
radium	88	Ra	[226]
actinium	89	Ac	[227]
thorium	90	Th	232.0
protactinium	91	Pa	231.0
uranium	92	U	238.0
neptunium	93	Np	237.0
plutonium	94	Pu	244.0
americium	95	Am	243.0
curium	96	Cm	247.0
berkelium	97	Bk	247.0
californium	98	Cf	251.0
einsteinium	99	Es	252.0
fermium	100	Fm	257.0
mendelevium	101	Md	258.0
nobelium	102	Nb	259.0
lawrencium	103	Lr	[262]
rutherfordium	104	Rf	[261]
bohrium	107	Bh	[262]
hassium	108	Hs	[265]
meitnerium	109	Mt	[266]
darmstadtium	110	Ds	[269]
roentgenium	111	Rg	[272]
unbihassium	112	Uub	[285]
unbinilium	114	Uuq	[289]



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MALAWI INSTITUTE OF TECHNOLOGY

APPLIED STUDIES DEPARTMENT

2018/2019 ACADEMIC YEAR

END OF SEMESTER II EXAMINATIONS

CHEM-I22: GENERAL CHEMISTRY II

DATE: 22ND JULY, 2019

TIME: 9:00 AM- 12:00 NOON

EXAMINERS: DR D. MWETA, DR B. THOLE, MS J. KUMATSO AND MS
L. BANDA

INSTRUCTIONS

1. This question paper contains five questions and six printed pages including the cover page, please check.
2. Answer all questions.
3. Answer each question on a fresh page of the answer sheet.
4. Marks for each question are indicated on the right margin.
5. All working must be clearly shown.
6. The periodic table is provided.
7. The following constants may be used:

$$R = 8.314 \text{ J/mol K}$$

$$= 0.08206 \text{ atm.L/mol.K}$$

$$1 \text{ atm} = 760 \text{ mmHg}$$

DO NOT TURN OVERLEAF WITHOUT INSTRUCTION

QUESTION ONE (18 marks)

- a. State the following:
- i. Charles' law [2]
 - ii. Dalton's law of partial pressures [2]
- b. A gas in a balloon occupies 2.5 L at 300 K, at what temperature will the balloon expand to 7.5 L? [3]
- c. Calculate the volume of H_2O that can be produced at 125°C and 815 mmHg when 74 g of C_2H_4 reacts with excess oxygen. [5]
- $$\text{C}_2\text{H}_4 (\text{g}) + 3\text{O}_2 (\text{g}) \longrightarrow 2\text{CO}_2 (\text{g}) + 2\text{H}_2\text{O} (\text{g})$$
- d. A rigid 4.5 L sealed vessel at 25°C contains 0.25 mol of nitrogen, 0.165 mol of argon and 0.1 mol of helium. The total pressure in the vessel is 1.75 atm. Calculate the partial pressure of argon. [6]

QUESTION TWO (18 marks)

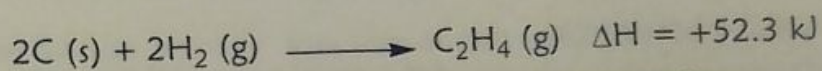
- a. Define the following terms:
- i. Heat [2]
 - ii. Standard enthalpy of formation [2]
- b. Given the thermochemical equation



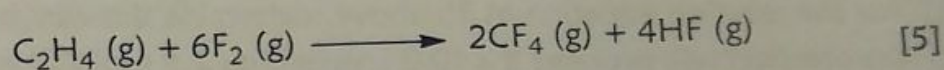
Calculate the heat (kJ) released to the surrounding when 12 g of $\text{CO} (\text{g})$ reacts completely. [4]

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c. From the following enthalpies of reaction:



Calculate the enthalpy of the following reaction below



d. Calculate the standard enthalpy of change for the following reaction: [5]



Substance	ΔH_f (kJ/mol)
$\text{SiCl}_4 (\text{l})$	-640.10
$\text{SiO}_2 (\text{s})$	-910.90
$\text{HCl} (\text{aq})$	-167.20
$\text{HCl} (\text{g})$	-92.30
$\text{H}_2\text{O} (\text{g})$	-241.82
$\text{H}_2\text{O} (\text{l})$	-285.83

QUESTION THREE (19 marks)

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a. State the Le Chatelier's principle. [2]

b. An important reaction in the commercial production of hydrogen is



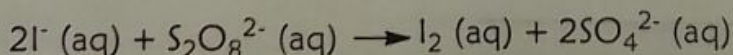
How will this system at equilibrium change (shift) in each of the following cases:

- i. Removing gaseous carbon monoxide. [2]
 - ii. Adding water vapour. [2]
 - iii. In a rigid reaction container, the pressure is increased by adding helium gas. [2]
 - iv. Decreasing the volume of the reaction container. [2]
- c. For the reaction equation in 3b above, determine the way in which the reaction will proceed if a 10.0 L vessel has 2.5 mol CO and H₂O, and 5.0 mol CO₂ and H₂ gas at 588 K ($K_c = 31.4$). [5]
- d. An equilibrium mixture in a 1.0 L container contains 0.25 mol PCl₅ and 0.16 mol of PCl₃. Calculate the equilibrium concentration of chlorine (Cl₂). [4]



QUESTION FOUR (25 marks)

- a. Explain what is meant by the term "rate of reaction". [2]
- b. Given the following equation and experimental data:

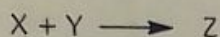


Run	[I ⁻] _o (M)	[S ₂ O ₈ ²⁻] _o (M)	Initial rates (M/s)
1	0.080	0.040	12.5 × 10 ⁻⁶
2	0.040	0.040	6.25 × 10 ⁻⁶
3	0.080	0.020	6.25 × 10 ⁻⁶
4	0.032	0.040	5.00 × 10 ⁻⁶
5	0.060	0.030	7.00 × 10 ⁻⁶

- i. Determine the order of reaction with respect to each reactant. [6]
- ii. Write the rate law for the reaction. [1]

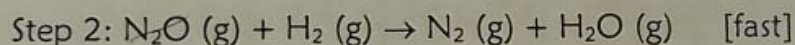
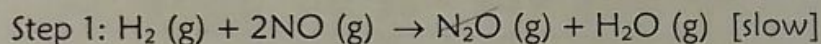
- iii. What is the overall order of reaction? [1]
 iv. Calculate the rate constant for the reaction. [3]

c. For the given reaction:



The activation energy for the above reaction equation is 39.4 kJ/mol. The rate constant at 215 °C is 5.0×10^{-3} . Calculate the rate constant at 452 °C. [6]

d. A proposed mechanism for the reduction of nitrogen as NO by hydrogen is as follows:



- i. What is the molecularity of step 1? [1]
 ii. Write the rate law for each elementary step? [2]
 iii. Write the overall equation for the reaction. [2]
 iv. Write the rate law for the reaction? [1]

QUESTION FIVE (20 marks)

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- a. Differentiate the following:
 i. Aliphatic and aromatic hydrocarbons [2]
 ii. Saturated and unsaturated hydrocarbons [2]
 b. Draw the structural formulas of the following:
 i. Trans-3-ethylhex-2-ene [2]
 ii. 4-ethyl-5-methylhept-2-yne [2]
 iii. m-dipropylbenzene [2]
 c. Draw and name all the isomers of hexane (C_6H_{14}) [10]

END OF QUESTION PAPER

The Periodic Table of the Elements									
Key:									
element name atomic number symbol atomic mass									
hydrogen 1 H 1.008	beryllium 4 Be 9.012	lithium 3 Li 6.941	sodium 11 Na 22.99	magnesium 12 Mg 24.31	calcium 20 Ca 40.08	potassium 19 K 39.10	rubidium 37 Rb 85.47	cesium 55 Cs 132.9	francium 87 Fr [223]
scandium 21 Sc 44.96	titanium 22 Ti 47.88	vanadium 23 V 50.94	chromium 24 Cr 52.00	manganese 25 Mn 54.94	iron 26 Fe 55.85	cobalt 27 Co 58.93	nickel 28 Ni 58.69	copper 29 Cu 63.55	zinc 30 Zn 65.39
yttrium 39 Y 88.91	zirconium 40 Zr 91.22	niobium 41 Nb 92.91	molybdenum 42 Mo 95.94	technetium 43 Tc [98]	ruthenium 44 Ru 101.1	rhodium 45 Rh 102.9	palladium 46 Pd 106.4	silver 47 Ag 107.9	cadmium 48 Cd 112.4
lutetium 71 Lu 175.0	hafnium 72 Hf 178.5	tantalum 73 Ta 180.9	tungsten 74 W 183.9	rhenium 75 Re 186.2	osmium 76 Os 190.2	iridium 77 Ir 192.2	platinum 78 Pt 195.1	gold 79 Au 197.0	mercury 80 Hg 200.6
barium 56 Ba 137.3	lanthanum 57 La 138.9	cerium 58 Ce 140.1	praseodymium 59 Pr 140.9	neodymium 60 Nd 144.2	promethium 61 Pm [145]	samarium 62 Sm 150.4	europium 63 Eu 152.0	gadolinium 64 Gd 157.3	terbium 65 Tb 158.9
strontium 38 Sr 87.62	yttrium 39 Y 88.91	zirconium 40 Zr 91.22	niobium 41 Nb 92.91	molybdenum 42 Mo 95.94	technetium 43 Tc [98]	rhodium 45 Rh 102.9	palladium 46 Pd 106.4	silver 47 Ag 107.9	cadmium 48 Cd 112.4
potassium 19 K 39.10	calcium 20 Ca 40.08	titanium 22 Ti 47.88	vanadium 23 V 50.94	chromium 24 Cr 52.00	manganese 25 Mn 54.94	iron 26 Fe 55.85	cobalt 27 Co 58.93	nickel 28 Ni 58.69	copper 29 Cu 63.55
rubidium 37 Rb 85.47	strontium 38 Sr 87.62	yttrium 39 Y 88.91	zirconium 40 Zr 91.22	niobium 41 Nb 92.91	molybdenum 42 Mo 95.94	technetium 43 Tc [98]	rhodium 45 Rh 102.9	palladium 46 Pd 106.4	silver 47 Ag 107.9
cesium 55 Cs 132.9	barium 56 Ba 137.3	lanthanum 57 La 138.9	cerium 58 Ce 140.1	praseodymium 59 Pr 140.9	neodymium 60 Nd 144.2	promethium 61 Pm [145]	samarium 62 Sm 150.4	europium 63 Eu 152.0	gadolinium 64 Gd 157.3
francium 87 Fr [223]	radium 88 Ra [226]	actinium 89 Ac [227]	thorium 90 Th 232.0	protactinium 91 Pa 231.0	uranium 92 U 238.0	neptunium 93 Np [237]	plutonium 94 Pu [244]	americium 95 Am [243]	curium 96 Cm [247]
beryllium 4 Be 9.012	lithium 3 Li 6.941	hydrogen 1 H 1.008	helium 2 He 4.003	boron 5 B 10.81	carbon 6 C 12.01	nitrogen 7 N 14.01	oxygen 8 O 16.00	fluorine 9 F 19.00	neon 10 Ne 20.18
aluminum 13 Al 26.98	silicon 14 Si 28.09	phosphorus 15 P 30.97	sulfur 16 S 32.07	chlorine 17 Cl 35.45	argon 18 Ar 39.95	potassium 19 K 39.10	calcium 20 Ca 40.08	scandium 21 Sc 44.96	titanium 22 Ti 47.88
gallium 31 Ga 69.72	germanium 32 Ge 72.59	arsenic 33 As 74.92	selenium 34 Se 78.96	bromine 35 Br 79.90	krypton 36 Kr 83.80	rubidium 37 Rb 85.47	strontium 38 Sr 87.62	yttrium 39 Y 88.91	zirconium 40 Zr 91.22
indium 49 In 114.8	tin 50 Sn 118.7	antimony 51 Sb 121.8	tellurium 52 Te 127.6	iodine 53 I 126.9	xenon 54 Xe 131.3	barium 56 Ba 137.3	lanthanum 57 La 138.9	cerium 58 Ce 140.1	praseodymium 59 Pr 140.9
thallium 81 Tl 204.4	lead 82 Pb 207.2	bismuth 83 Bi 209.0	polonium 84 Po [210]	astatine 85 At [210]	radon 86 Rn [222]	cesium 55 Cs 132.9	barium 56 Ba 137.3	lanthanum 57 La 138.9	cerium 58 Ce 140.1
unnilbium 114 Uub [285]	ununbium 115 Uub [285]	ununium 116 Uuq [289]	ununium 117 Uuq [289]	ununium 118 Uuq [289]	ununium 119 Uuq [289]	ununium 120 Uuq [289]	ununium 121 Uuq [289]	ununium 122 Uuq [289]	ununium 123 Uuq [289]

lanthanum	57	La	138.9	actinium	89	Ac	[227]
cerium	58	Ce	140.1	thorium	90	Th	232.0
praseodymium	59	Pr	140.9	protactinium	91	Pa	231
neodymium	60	Nd	144.2	uranium	92	U	238.0
promethium	61	Pm	[147]	neptunium	93	Np	[237]
samarium	62	Sm	150.4	plutonium	94	Pu	[242]
europtium	63	Eu	152.0	americium	95	Am	[243]
gadolinium	64	Gd	157.3	curium	96	Cm	[247]
terbium	65	Tb	158.9	berkelium	97	Bk	[247]
dysprosium	66	Dy	162.5	californium	98	Cf	[249]
holmium	67	Ho	164.9	einsteinium	99	Es	[254]
erbium	68	Er	167.3	fermium	100	Fm	[253]
thulium	69	Tm	168.9	mendelevium	101	Md	[256]
yterbium	70	Yb	173.0	nobelium	102	No	[254]

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MALAWI INSTITUTE OF TECHNOLOGY

APPLIED STUDIES DEPARTMENT

2019/2020 END OF SEMESTER TWO EXAMINATION

CHEM-I22: GENERAL CHEMISTRY II

DATE: 21ST DECEMBER, 2020

TIME: 9:00-12:00 NOON

EXAMINERS: DR D. MWETA, DR B. THOLE, MS J. KUMATSO A.
MTEWA AND P. CHIMTALI

INSTRUCTIONS

1. This question paper contains five questions and eight printed pages including the cover page, please check.
2. Answer each question on a fresh page of the answer sheet.
3. Marks for each question are indicated on the right margin.
4. All working must be clearly shown.
5. Answer all questions.
6. The periodic table is provided.
7. The following constants may be used:

$$R = 8.314 \text{ J/mol K or}$$

$$= 0.08206 \text{ atm.L/mol.K}$$

$$1 \text{ atm} = 760 \text{ mmHg}$$

DO NOT TURN OVERLEAF WITHOUT INSTRUCTION

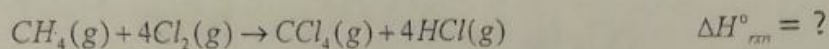
QUESTION ONE (18 marks)

- a. State the following: [2]
i. Gay Lussacs' law [2]
ii. Dalton's law of partial pressures
- b. A sample of a gas is in a container with a constant volume at 75 °C and at a pressure of 5.5 atm. Calculate the temperature of the gas if the pressure goes up to 10 atm. Give your answer in degrees Celsius. [6]
- c. A tank contains 52.5 g of oxygen and 65.1 g of CO₂ at 300 K. The total pressure in the tank is 9.21 atm. Calculate the partial pressure of each gas in the mixture. [8]

QUESTION TWO (20 marks)

- a. Define the following terms: [2]
i. System [2]
ii. Surrounding
- b. An automobile engine does 520 kJ of work and loses 220 kJ of energy as heat. What is the change in the internal energy of the engine? Treat the engine, fuel, and exhaust gases as a closed system. [4]
- c. A certain mass of water was heated with 41,840 Joules of energy, raising its temperature from 22.0 to 28.5 °C. Find the mass of the water, in grams (specific heat capacity of H₂O = 4.18 J/g °C). [3]
- d. How much work in joules is done on the system when a 1.15 atm external pressure causes a piston to decrease in volume from 6.55 L to 3.16 L? [4]

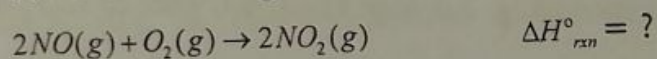
- e. Use the information provided to determine $\Delta H^\circ_{\text{rxn}}$ for the following reaction:



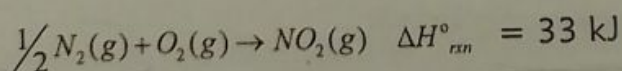
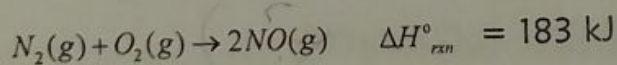
Substance	$\Delta H_f^\circ (\text{kJ/mol})$
$\text{CH}_4(\text{g})$	-75
$\text{CCl}_4(\text{g})$	-96
$\text{HCl}(\text{g})$	-92

[5]

- f. Use the standard reaction enthalpies given below to determine $\Delta H^\circ_{\text{rxn}}$ for the following reaction:



Given:

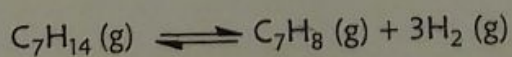


QUESTION THREE (20 marks)

- a. State the Le Chatelier's principle.

[2]

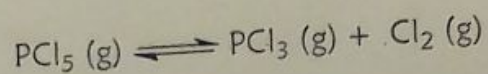
- b. Toluene, C_7H_8 , is an important organic solvent. It is made industrially from methyl cyclohexane.



The forward reaction is endothermic. Mention three changes to this equilibrium mixture that would shift the equilibrium towards more production of toluene (C_7H_8).

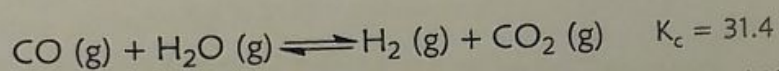
[6]

- c. The following reaction took place in a sealed flask at 250 °C.



At equilibrium, the gases in the flask had the following concentrations: $[\text{PCl}_5] = 1.2 \times 10^{-2} \text{ mol/L}$, $[\text{PCl}_3] = 1.5 \times 10^{-2} \text{ mol/L}$ and $[\text{Cl}_2] = 1.5 \times 10^{-2} \text{ mol/L}$. Calculate the value of K_c at 250 °C. [5]

- d. Carbon monoxide can react with steam under the appropriate conditions according to the equation:



Determine the way in which the reaction will proceed if a 10.0 L vessel has 2.5 mol CO , 2.5 mol H_2O , 5.0 mol of CO_2 and 5.0 mol H_2 at 588 K. [7]

QUESTION FOUR (27 marks)

- a. Define the following terms:

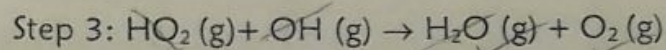
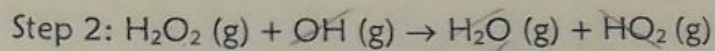
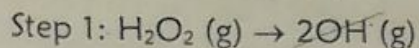
i. Reaction mechanism [2]

ii. Reaction intermediate [2]

iii. Elementary steps [2]

- b. State the three ways through which the catalyst increases the rate of a reaction. [3]

- c. A proposed mechanism for the decomposition of hydrogen peroxide consists of three elementary steps:



If the observed rate law for the reaction is first order with respect to H_2O_2 (i.e. $\text{Rate} = k[\text{H}_2\text{O}_2]$),

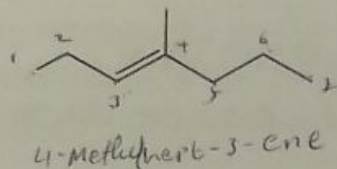
- Determine the molecularity of each elementary step. [3]
- Write the rate law for each elementary step. [6]
- Which step in the mechanism is the rate-determining step? Explain. [3]
$$2\text{H}_2\text{O}_2 (\text{g}) \rightarrow 2\text{H}_2\text{O} + \text{O}_2$$
- Identify the intermediate(s) in the reaction mechanism. [2]
- Write the overall reaction for the decomposition of hydrogen peroxide. [4]

uni/ monomolecular
bi molecular
termolecular

QUESTION FIVE (15 marks)

a. Name the following compounds using their correct IUPAC names

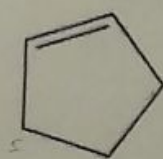
i.



terminal
terminal
terminal

[2]

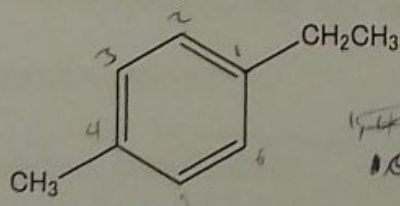
ii.



Cyclopentene

[2]

iii.



ethyl-4-methylcyclohexene

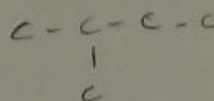
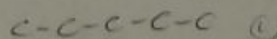
2-methylbenzene

2,2-dimethyl

[2]

b. Draw all the isomers of C_5H_{12}

alkane



[3]

c. Name the structural isomers in 5 b.

C_nH_{2n+2}
 C_5H_{12}

[6]

END OF QUESTION PAPER

2

Ne	20.18
Argon	39.95

[illegible]



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APPLIED STUDIES DEPARTMENT

END OF SEMESTER TWO EXAMINATIONS

PHYS 122: ELECTRICITY, MAGNETISM, WAVES AND VIBRATIONS

DATE: 10th August, 2018

TIME: 3 Hours (9:00 – 12:00)

Examiners: R. Chimombo & P. Mnjeza

INSTRUCTIONS

1. Write your **Examination Number** on each page of your answer sheets.
2. Start each question on a fresh page
3. This paper contains **7 printed pages** and **SIX QUESTIONS**, please check.
4. Answer **ALL** questions.
5. Marks allocated for each question are indicated against it.

You may not start to read the questions on the subsequent pages of this question paper until instructed to do so by the invigilator.

CONSTANT(S)

Coulomb constant	$k_e = 8.9875 \times 10^9 \text{ N.m}^2/\text{C}^2$
Permittivity of free space	$\epsilon_0 = 8.8542 \times 10^{-12} \text{ C}^2/\text{N.m}^2$
Magnitude of charge (Proton or electron)	$e = 1.6 \times 10^{-19} \text{ C}$
Take acceleration due to gravity	$g = 9.8 \text{ m/s}^2$
Mass of electron	$m_e = 9.1095 \times 10^{-31} \text{ Kg}$
Mass of proton	$m_p = 1.67261 \times 10^{-27} \text{ Kg}$
Permeability of free space	$\mu_0 = 4\pi \times 10^{-7} \text{ T.m/A}$

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QUESTION ONE:

- a. State Coulomb's law [2]
- b. Three charged particles are arranged in a line, as shown in **figure 1**.

Calculate the net electric force on particle 3 (the $-4.0 \mu\text{C}$ on the right) due to the other two charges. [4]

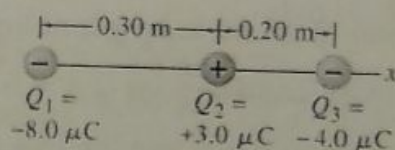


Figure 1

- c. What is the magnitude of the acceleration experienced by an electron in an electric field of 750 N/C ? [4]
- d. Convert 0.01 eV to joules [2]
- e. A proton is released from rest in a uniform electric field that has a magnitude of $8.0 \times 10^4 \text{ V/m}$. The proton undergoes a displacement of 0.50 m in the direction of $\mathbf{E}$.
- Find the change in the electric potential for this displacement. [2]
 - Find the change in potential energy of the field-system for this displacement [2]
 - Find the speed of the proton after completing the 0.50 m displacement in the electric field. [4]

QUESTION TWO:

- a. A hair dryer draws 7.5 A when plugged into a 120 V line.
- What is its resistance? [2]
 - How much charge passes through it in 15 min? (Assume direct current) [3]
- b. What is the diameter of a 1.00 m length of tungsten wire whose resistance is 0.35Ω ? Resistivity of tungsten wire is $\rho = 5.6 \times 10^{-8} \Omega \cdot \text{m}$. [3]
- c. If the EMF of the battery is 9 V and the internal resistance of the battery, $r = 2 \Omega$, **figure 2**. Find the voltmeter and ammeter readings when **not connected** to anything. [2]

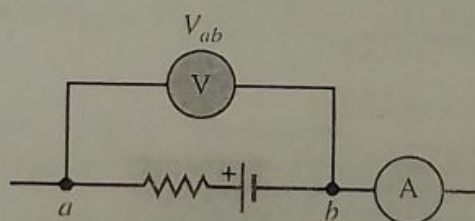


Figure 2

- d. Consider the circuit diagram shown in **figure 3**.
- Find the equivalent resistance in the circuit. [3]
 - Find the current in the 470Ω resistor. [1]

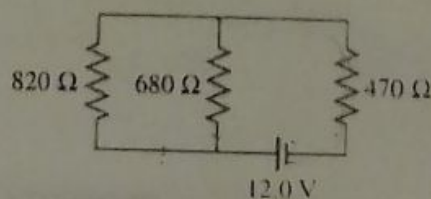


Figure 3

- e. For the circuit shown in **figure 4**, calculate the current in the $2.00\ \Omega$ resistor.

[6]

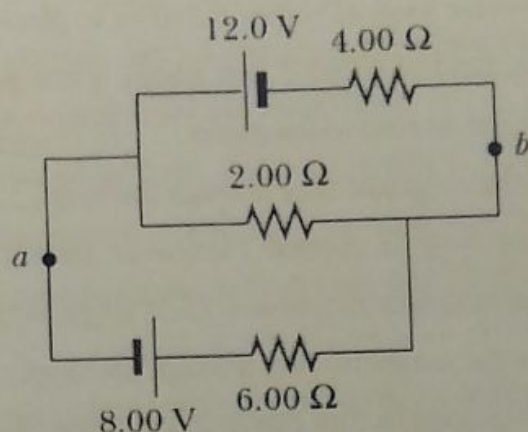


Figure 4

QUESTION THREE:

- a. A wire having a mass per unit length of 0.500 g/cm carries a 2.00-A current horizontally to the south. What are the direction and magnitude of the minimum magnetic field needed to lift this wire vertically upward? [5]
- b. Alpha particles of charge $q = +2e$ (where $e = 1.6 \times 10^{-19}\text{ C}$) and mass $m = 6.6 \times 10^{-27}\text{ kg}$ are emitted from a radioactive source at a speed of $1.6 \times 10^7\text{ m/s}$. What magnetic field strength would be required to bend them into a circular path of radius $r = 0.25\text{ m}$? [5]
- c. The magnitude of the magnetic field at 4.0 cm from an infinitely long, straight current-carrying wire is $2.5 \times 10^{-5}\text{ T}$. Calculate the magnitude of the current in the wire. [4]

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d. A conducting circular loop of radius 0.250 m is placed in the xy -plane in a uniform magnetic field of 0.360 T that points in the positive z -direction, the same direction as the normal to the plane

- i. Calculate the magnetic flux through the loop. [3]
- ii. Suppose the loop is rotated clockwise around the x -axis, so the normal direction now points at a 45.0° angle with respect to the z -axis. Recalculate the magnetic flux through the loop. [2]
- iii. What is the change in flux due to the rotation of the loop? [1]

QUESTION FOUR:

- a. Show that the capacitance of a charged sphere is given by $4\pi\epsilon_0 R$. [4]
- b. The electric field between the plates of a paper-separated ($k = 3.75$) capacitor is 8.24×10^4 V/m. the plates are 1.95 mm apart, and the charge on each plate is $0.775 \mu\text{C}$. Determine:
 - i. The capacitance of this capacitor. [3]
 - ii. The area of each plate. [3]

QUESTION FIVE:

- a. An object is located 1.5 m from an 8.0 D lens.
 - i. Find the image distance. [5]
 - ii. Find the magnification M . [2]
 - iii. Describe the nature of the image. [3]

QUESTION SIX

- a. At what displacement from equilibrium is the speed of a simple harmonic motion half the maximum value? [4]
- b. Derive the formula for the maximum speed $v_{\max}$ of a simple pendulum bob in terms of g , length L and the angle of swing θ . [4]

- c. White light passes through two slits 0.50 mm apart, and an interference pattern is observed on a screen 2.5 m away. The first-order fringe resembles a rainbow with violet and red light at opposite ends. The violet light falls about 2.0 mm and the red 3.4 mm from the center of the central white fringe. Estimate the wavelength for the violet light. [3]
- d. A glass tube (open at both ends) of length L is positioned near an audio speaker of frequency $f = 680$ Hz. For what smallest ($n = 1$) possible value of L will the tube resonate with the speaker? Take 343 m/s as the speed of sound in air. [3]
- e. In searching the bottom of a pool at night, a watchman shines a narrow beam of light from his flashlight, 1.3 m above the water level, onto the surface of the water at a point 2.7 m from the edge of the pool, **figure 6**. Where does the spot of light hit the bottom of the pool, measured from the wall beneath his foot, if the pool is 2.1 m deep?(index of refraction of air $n_1 = 1.00$ and index of refraction of water $n_2 = 1.33$) [6]

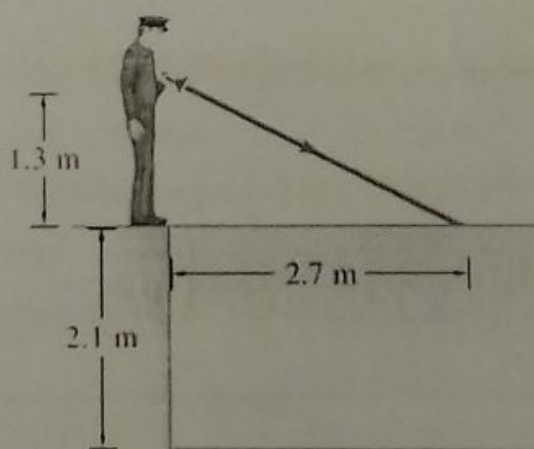


Figure 5

END OF QUESTIONS

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APPLIED STUDIES DEPARTMENT

2018/19 END OF SEMESTER TWO EXAMINATIONS

PHYS 122: ELECTRICITY AND MAGNETISM, VIBRATIONS AND WAVES

DATE: 29th July 2019.

TIME: 3 Hours (9:00 -12:00)

Examiners: R. Chimombo, L. Chisanu, JSP Mlatho, B. Mmame & P. Mnjeza.

INSTRUCTIONS

1. Write your Examination Number on each page of your answer sheets.
 2. Start each question on a fresh page.
 3. This paper contains Nine (9) printed pages and five (5) QUESTIONS, please check.
 4. Answer ALL questions.
 5. Marks allocated for each question are indicated against it.
 6. Round off the final answer to 2 decimal places.
-

SOME USEFUL CONSTANT(S)

Acceleration due to gravity

$$g = 9.8 \text{ m/s}^2$$

Permeability of free space

$$\mu_0 = 4\pi \times 10^{-7} \text{ T.m/A}$$

Magnitude of charge (proton or electron)

$$e = \pm 1.60 \times 10^{-19} \text{ C}$$

Permittivity of free space

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ C}^2/\text{N.m}^2$$

Coulomb's Constant

$$k_e = 8.99 \times 10^9 \text{ N.m}^2/\text{C}^2$$

Mass of a proton

$$m_p = 1.67 \times 10^{-27} \text{ kg}$$

Mass of an electron

$$m_e = 9.11 \times 10^{-31} \text{ kg}$$

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Question 1

- a. A hair dryer draws 7.5 A when plugged into a 120 V line.
- What is its resistance? [2]
 - How much charge passes through it in 15 min? (Assume direct current). [3]
- b. What is the diameter of a 1.00 m length of tungsten wire whose resistance is $0.35\ \Omega$? Resistivity of tungsten wire, $\rho = 5.6 \times 10^{-8}\ \Omega \cdot \text{m}$. [3]
- c. Consider the circuit diagram shown in figure 1. Find the equivalent resistance in the circuit. [4]

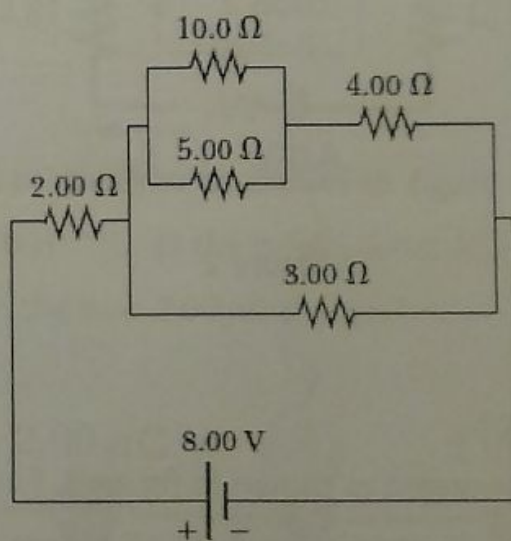


Figure 1

- d. Determine the currents I_1 , I_2 and I_3 in figure 2. Assume the internal resistance of each battery is $r = 1.00\Omega$ [8]

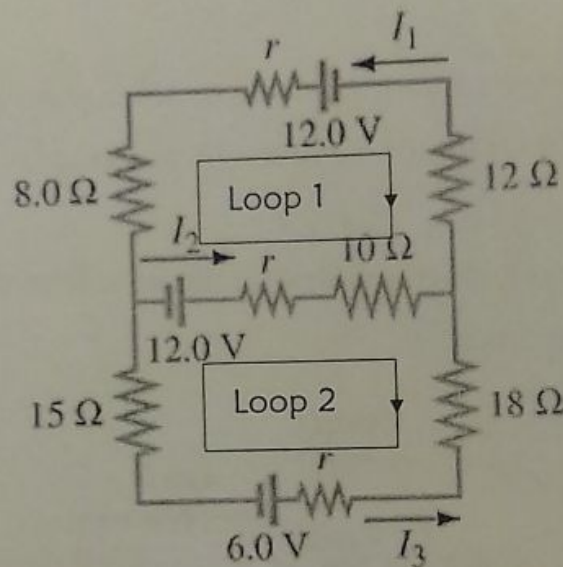


Figure 2

Question 2

- a. Define electric charge. [2]
- b. Three charges are arranged as shown in figure 3. Find the magnitude and direction of the resultant force on 6.00 nC charge due to 5.00 nC and -3.00 nC charges. [8]

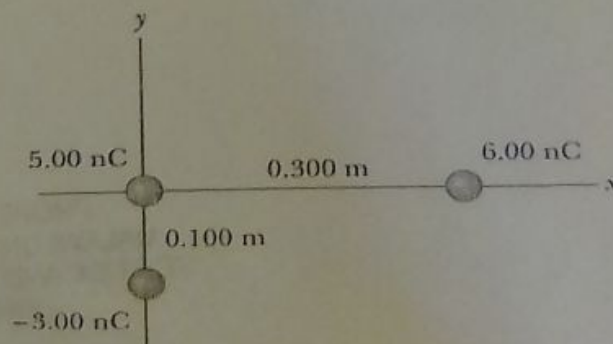


Figure 3

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- c. Two identical small charged spheres, each having a mass of $3.0 \times 10^{-2} \text{ kg}$, hang in equilibrium as shown in Figure 4. The length of each string is 0.15 m and the angle is $\theta = 5.0^\circ$. Find the magnitude of the charge on each sphere. [10]

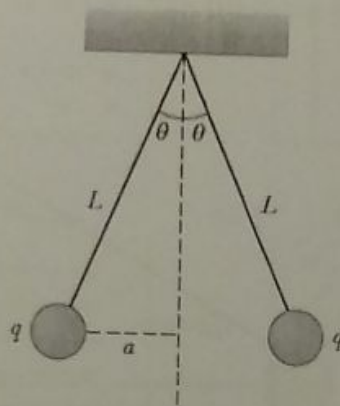


Figure 4

Question 3

- a. Given two $2.00 \mu\text{C}$ charges, as shown in figure 5 and a positive test charge $q = 1.28 \times 10^{-8} \text{ C}$ at the origin, what is the electric potential at the origin due to the two $2.00 \mu\text{C}$ charges? [4]

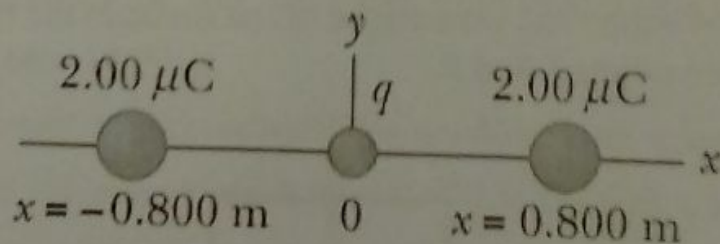


Figure 5

- b. How much work is required to assemble three point charges, at the corners of a right-angled triangle as in figure 6 where $a = 0.300 \text{ m}$, $b = 0.400 \text{ m}$, and $q = 5.00 \mu\text{C}$. [6]

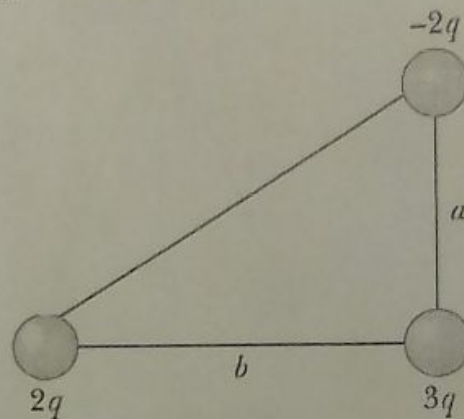


Figure 6

- c. Two conductors having net charges of $+10.0 \mu\text{C}$ and $-10.0 \mu\text{C}$ have a potential difference of 10.0 V between them. Determine the capacitance of the system. [3]
- d. Find the equivalent capacitance of the circuit in the figure 7 below between points A and B. [7]

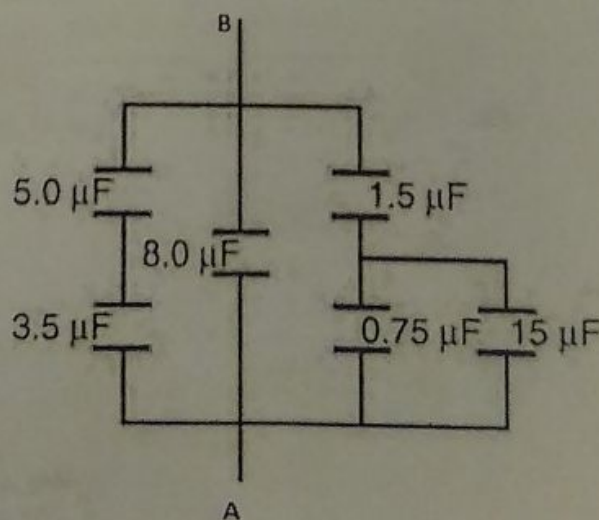


Figure 7

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Question 4

- a. Determine the initial direction of the deflection of charged particles as they enter the magnetic fields as shown in figure 8. [4]

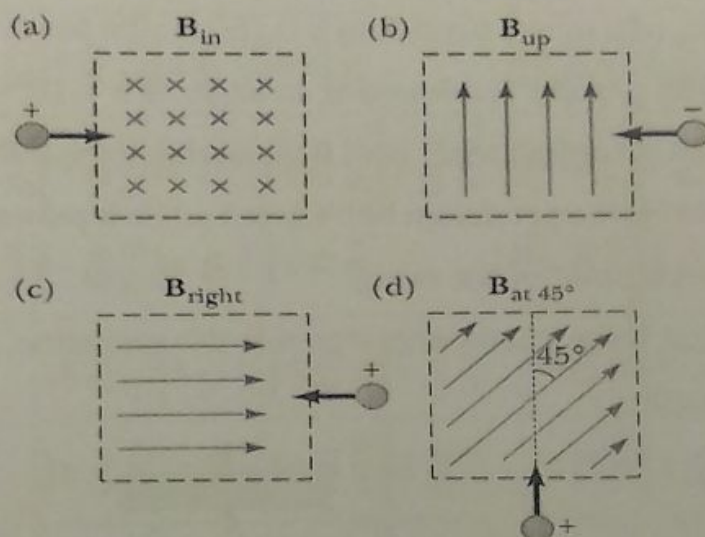


Figure 8

- b. A singly charged positive ion moving at $4.60 \times 10^5 \text{ m/s}$ leaves a circular track of radius 7.94 mm along a direction perpendicular to the 1.80 T magnetic field of a bubble chamber. Compute the mass (in atomic mass units) of this ion, and from that value, identify it. Hint: $1\text{u} = 1.66 \times 10^{-27} \text{ kg}$. [6]
- c. A singly charged ion completes five revolutions in a uniform magnetic field of magnitude $5.00 \times 10^{-2} \text{ T}$ in 1.50 ms. Calculate the mass of the ion in kilograms. [5]
- d. A proton having a kinetic energy of 5.00 MeV is moving in the positive x-direction and enters a magnetic field $B = 0.050 \text{ T}$ directed out of the plane of the page. Calculate the momentum of the proton as it leaves the magnetic field. ($1\text{eV} = 1.60 \times 10^{-19} \text{ J}$). [5]

Question 5

- a. The length of a simple pendulum is 0.760 m, the pendulum bob has a mass of 365 g, and it is released at an angle of $\theta = 12^\circ$ to the vertical.
- With what frequency does it vibrate? Assume SHM. [3]
 - What is the pendulum bob's speed when it passes through the lowest point of the swing? [6]
 - What is the total energy stored in this oscillation, assuming no losses? [3]

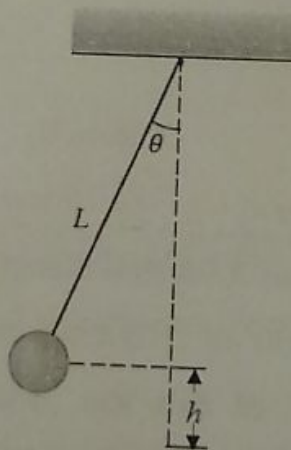


Figure 9

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- b. An object with mass of 3.00 kg is attached to a spring with spring stiffness constant $k = 280 \text{ N/m}$ and executing a simple harmonic motion. When the object is 0.020 m from its equilibrium position, it is moving with a speed of 0.55 m/s.
- Calculate the amplitude of the motion [5]
 - Calculate the maximum velocity attained by the object. [3]

END OF QUESTIONS

The Periodic Table of the Elements

hydrogen 1 H		beryllium 4 Be		lithium 3 Li		sodium 11 Na		potassium 19 K		rubidium 37 Rb		cesium 55 Cs		francium 87 Fr	
1.008	9.012	6.941	22.99	39.10	85.47	132.9	22.99	39.10	85.47	132.9	22.99	39.10	85.47	132.9	22.99
Key: element name atomic number symbol atomic mass															
scandium 21 Sc		titanium 22 Ti		vanadium 23 V		chromium 24 Cr		manganese 25 Mn		iron 26 Fe		cobalt 27 Co		nickel 28 Ni	
44.96	47.88	50.94	52.00	54.94	55.85	58.93	58.69	63.55	65.39	68.93	72.64	74.92	78.97	79.90	83.80
yttrium 39 Y	zirconium 40 Zr	niobium 41 Nb	molybdenum 42 Mo	technetium 43 Tc	ruthenium 44 Ru	rhodium 45 Rh	palladium 46 Pd	silver 47 Ag	cadmium 48 Cd	indium 49 In	tin 50 Sn	antimony 51 Sb	tellurium 52 Te	iodine 53 I	xenon 54 Xe
88.91	91.22	92.91	95.94	[98]	101.1	102.9	106.4	107.9	112.4	114.8	118.7	121.8	127.6	126.9	131.3
lutetium 71 Lu	hafnium 72 Hf	tantalum 73 Ta	tungsten 74 W	rhenium 75 Re	osmium 76 Os	iridium 77 Ir	platinum 78 Pt	gold 79 Au	mercury 80 Hg	thallium 81 Tl	lead 82 Pb	bismuth 83 Bi	polonium 84 Po	astatine 85 At	radon 86 Rn
175.0	178.5	180.9	183.9	186.2	190.2	192.2	195.1	197.0	200.6	204.4	207.2	209.0	[210]	[210]	[222]
lawrencium 103 Lr	rutherfordium 104 Rf	dubnium 105 Db	seaborgium 106 Sg	bohrium 107 Bh	hassium 108 Hs	meitnerium 109 Mt	darmstadtium 110 Ds	roentgenium 111 Rg	unbinilium 112 Uub	unquadrium 114 Uuq	unseptilium 115 Uus	unhexilium 116 Uuh	unnilium 117 Uun	untridecium 118 Uut	unquadium 119 Uuq

[illegible]

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APPLIED STUDIES DEPARTMENT

2019/20 END SEMESTER EXAMINATIONS

PHYS 122: ELECTRICITY AND MAGNETISM, WAVES AND VIBRATIONS

[100 Marks]

Date: 14th December, 2020.

TIME: 3 Hours (9:00 – 12:00)

Examiners: F. Benard, R. Chimombo, S. Chisale, C. Daka, C. Malanda & P. Mnjeza.

INSTRUCTIONS

1. Write your Examination Number on each page of your answer sheets.
2. Start each question on a fresh page
3. This paper contains 7 pages and 6 QUESTIONS, please check.
4. Answer ALL questions.
5. Marks allocated for each question are indicated against it.
6. Show your work clearly.
7. Round off your final answer to 2 decimal places.

[You may not start to read the questions printed on subsequent pages on this question paper until instructed to do so by the invigilator]

CONSTANT(S)

Coulomb constant	$k_e = 8.9875 \times 10^9 \text{ N.m}^2/\text{C}^2$
Permittivity of free space	$\epsilon_o = 8.8542 \times 10^{-12} \text{ C}^2/\text{N.m}^2$
Magnitude of charge (Proton or electron)	$e = 1.6 \times 10^{-19} \text{ C}$
Take acceleration due to gravity	$g = 9.8 \text{ m/s}^2$
Mass of electron	$m_e = 9.1095 \times 10^{-31} \text{ Kg}$
Mass of proton	$m_p = 1.67261 \times 10^{-27} \text{ Kg}$
Permeability of free space	$\mu_o = 4\pi \times 10^{-7} \text{ T.m/A}$

12.566
0000001256

Question 1

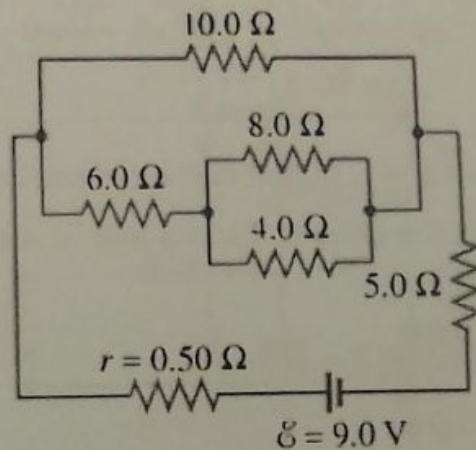
a. Define Current.

[2 marks]

b. A 9.0 V battery whose internal resistance r is 0.50Ω is connected in the circuit shown in fig. 2. What is the current in the 8.0Ω resistor?

[9 marks]

$R_{eq} = 10$
 $V = 8.56$
 $I = \frac{V}{R} = 0.856$



$I = 0.856$
 $V = 8.56$
 $R = \frac{V}{I} = 0.204 V$
 0.034
 8.356
 $V = 1R$
 $I = \frac{V}{R}$

Figure 1

c. For the circuit shown in figure 1. Find all the currents in the circuit

[9 marks]

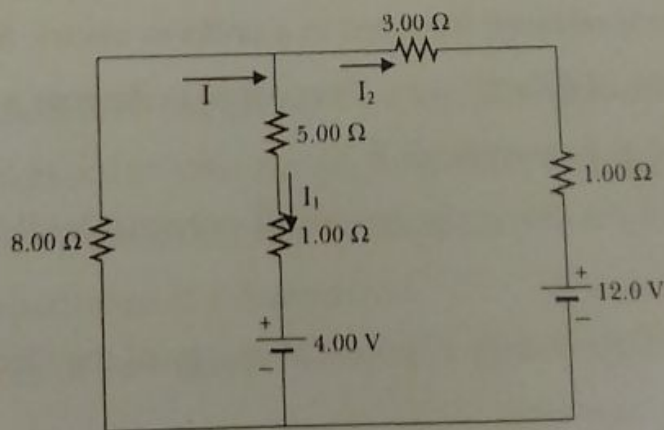


Figure 2

$R = \frac{V}{I}$

Question 2

- a. Three charged particles are arranged in a line, as shown in figure 3. Calculate the net electric force on particle 3 (the $-4.0 \mu\text{C}$ on the right) due to the other two charges. [6 marks]

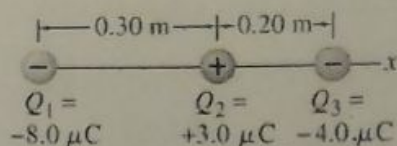


Figure 3

- b. Consider fig. 7. Let $Q = 3.00 \mu\text{C}$, find the direction and magnitude of the electric field at the point P. ($a = 2 \text{ mm}$, $x = 6.00 \text{ mm}$,) [9 marks]

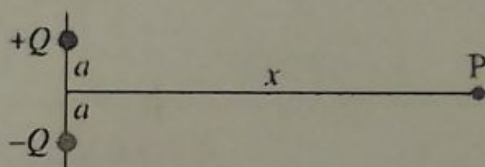


Figure 4

Question 3

- a. A proton is released from rest in a uniform electric field that has a magnitude of $8.0 \times 10^4 \text{ V/m}$. The proton undergoes a displacement of 0.50 m in the direction of $\mathbf{E}$.
- Find the change in the electric potential for this displacement. [3 marks]
 - Find the change in potential energy of the field-system for this displacement [3 marks]
 - Find the speed of the proton after completing the 0.50 m displacement in the electric field. [4 marks]



$$\begin{aligned}
 U &= -qEx \\
 &= qE(x_1 - x_2) \\
 &= 6.4 \times 10^{-15}
 \end{aligned}$$

- b. Three negative point charges lie along a line as shown in figure 5. Find the magnitude of the electric potential (ΔV) this combination of charges produces at point P , which lies 6.00 cm from the $-2.00 \mu\text{C}$ charge measured perpendicular to the line connecting the three charges. [5 marks]

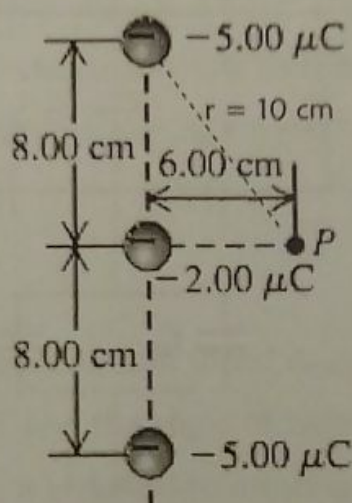


Figure 5

$$I = \frac{V}{R}$$

$$V = IR$$

0.856

Question 4

- a. Show that the capacitance of a charged sphere is given by $4\pi\epsilon_0 R$.

[4 marks]

- b. The electric field between the plates of a paper-separated ($k = 3.75$) capacitor is $8.24 \times 10^4 \text{ V/m}$. the plates are 1.95 mm apart, and the charge on each plate is $0.775 \mu\text{C}$. Determine:

- i. The capacitance of this capacitor.

[3 marks]

- ii. The area of each plate.

[3 marks]

$$E = \frac{\Delta V}{d}$$

$$\frac{1.9}{k}$$

8.56

10

$$I = \frac{V}{R}$$

$$C = \frac{Q}{\Delta V}$$

$$I = \frac{V}{R}$$

0.204

$V = 6 \times 0.204$

0.034

0.856

$V = IR$

8.356

- c. Find the equivalent capacitance between points a and b for the group of capacitors connected as shown in Figure 6. Take $C_1 = 5.00 \mu F$, $C_2 = 10.0 \mu F$ and $C_3 = 2.00 \mu F$. [5 marks]

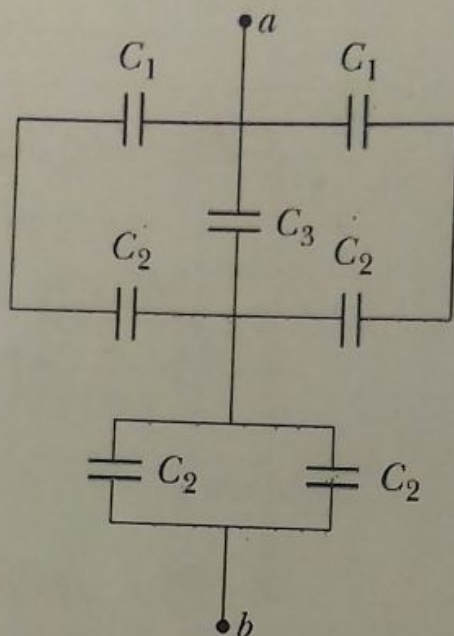


Figure 6

5.5
 $V = 0.89$
 $V = IR$

Question 5

- a. Alpha particles of charge $q = +2e$ (where $e = 1.6 \times 10^{-19} C$) and mass $m = 6.6 \times 10^{-27} kg$ are emitted from a radioactive source at a speed of $1.6 \times 10^7 m/s$. What magnetic field strength would be required to bend them into a circular path of radius $r = 0.25 m$? [5 marks]
- b. The magnitude of the magnetic field at $4.0 cm$ from an infinitely long, straight current-carrying wire is $2.5 \times 10^{-5} T$. Calculate the magnitude of the current in the wire. [4 marks]

$$V^2 = \sqrt{\frac{k}{m} (A^2 - x^2)^2}$$

$$V^2 \times m = \frac{k(A^2 - x^2)}{m V^2} \times m$$

$$\frac{mv^2}{r} = qvB \sin \theta$$

- c. A conducting circular loop of radius 0.250 m is placed in the xy -plane in a uniform magnetic field of 0.360 T that points in the positive z -direction, the same direction as the normal to the plane
- Calculate the magnetic flux through the loop. [6 marks]
 - Suppose the loop is rotated clockwise around the x -axis, so the normal direction now points at a 45.0° angle with respect to the z -axis. Recalculate the magnetic flux through the loop. [3 marks]
 - What is the change in flux due to the rotation of the loop? [2 mark]

Question 6

- Derive the formula for the maximum speed v_{max} of a simple pendulum bob in terms of g , length L and the angle of swing θ . [7 marks]
- A block of unknown mass is attached to a spring with a spring constant of 6.50 N/m and undergoes simple harmonic motion with an amplitude of 10.0 cm. When the block is halfway between its equilibrium position and the end point, its speed is measured to be 30.0 cm/s. Calculate the mass of the block. [8 marks]

$$\frac{1}{2}mv^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

$$kx^2 = mv^2 + kx^2$$

END OF QUESTIONS

$$\frac{1}{2}mv^2 = \frac{1}{2}mgh^2$$

$$\frac{1}{2}mv^2 = mgh + \frac{1}{2}mv^2$$

$$\frac{1}{2}mgL^2 = \frac{1}{2}mg$$

$$\frac{1}{2}(gL^2) = \frac{1}{2}mgh + \frac{1}{2}mv^2$$

$$mgL^2 = mgh + mv^2$$

$$gL^2 - mgh = \frac{mv^2}{m}$$

$$\frac{g}{m} \left(\frac{L^2}{h} \right) = v$$

$$V_{max} = \sqrt{\frac{g}{L} (A^2)}$$

V_{max} PE 0

$$\frac{1}{2}mv_i^2 + \frac{1}{2}kx_i^2 = \frac{1}{2}mv_f^2 + \frac{1}{2}kx_f^2$$

$$\frac{1}{2}mv_i^2 + \frac{1}{2}mgh_i = \frac{1}{2}mv_f^2 + \frac{1}{2}mgh_f$$

$$\frac{1}{2}mgh_i = \frac{1}{2}mv_f^2$$

$$v^2 = gh$$

$$v = \sqrt{gh}$$

$$\frac{g}{L} (A^2)$$

- c. A conducting circular loop of radius 0.250 m is placed in the xy -plane in a uniform magnetic field of 0.360 T that points in the positive z -direction, the same direction as the normal to the plane
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$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

$$\frac{1}{2}kx^2 = mv^2 + kx^2$$

END OF QUESTIONS

$$\frac{1}{2}mv^2 = \frac{1}{2}mgh^2$$

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$$\frac{1}{2}mgL^2 = \frac{1}{2}mg$$

$$\frac{1}{2}(gL^2) = \frac{1}{2}mgh + \frac{1}{2}mv^2$$

$$mgL^2 = mgh + mv^2$$

$$gL^2 - mgh = \frac{mv^2}{m}$$

$$\sqrt{\frac{g}{m} \left(\frac{L^2}{h} \right)} = v$$

$$V_{max} = \sqrt{\frac{g}{L} (A^2)}$$

$$V_{max} \quad PE = 0$$

$$\frac{1}{2}mv_i^2 + \frac{1}{2}kx_i^2 = \frac{1}{2}mv_f^2 + \frac{1}{2}kx_f^2$$

$$\frac{1}{2}mv_i^2 + \frac{1}{2}mgh_i = \frac{1}{2}mv_f^2 + \frac{1}{2}mgh_f$$

$$\frac{1}{2}mgh_i = \frac{1}{2}mv_f^2$$

$$v^2 = gh$$

$$v = \sqrt{gh}$$

c. A conducting circular loop of radius 0.250 m is placed in the xy -plane in a uniform magnetic field of 0.360 T that points in the positive z -direction, the same direction as the normal to the plane

- Calculate the magnetic flux through the loop. [6 marks]
- Suppose the loop is rotated clockwise around the x -axis, so the normal direction now points at a 45.0° angle with respect to the z -axis. Recalculate the magnetic flux through the loop. [3 marks]
- What is the change in flux due to the rotation of the loop? [2 mark]

Question 6

a. Derive the formula for the maximum speed v_{max} of a simple pendulum bob in terms of g , length L and the angle of swing θ . [7 marks]

b. A block of unknown mass is attached to a spring with a spring constant of 6.50 N/m and undergoes simple harmonic motion with an amplitude of 10.0 cm. When the block is halfway between its equilibrium position and the end point, its speed is measured to be 30.0 cm/s. Calculate the mass of the block. [8 marks]

$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

$$\boxed{\frac{1}{2}kx^2} = mv^2 + kx^2$$

END OF QUESTIONS

$$\frac{1}{2}mv^2 = \frac{1}{2}mgh^2$$

$$\frac{1}{2}mv^2 = mgh + \frac{1}{2}mv_f^2$$

$$\frac{1}{2}kA^2 = \frac{1}{2}mg$$

$$\frac{1}{2}(gL^2) = \frac{1}{2}mgh + \frac{1}{2}mv^2$$

$$mgh^2 = mgh + mv^2$$

$$gL^2 - mgh = \frac{mv^2}{m}$$

$$\sqrt{\frac{g}{m} \left(\frac{L^2}{h} \right)} = v$$

$$V_{max} = \sqrt{\frac{2}{L} (A^2)}$$

V_{max} PE 0

$$\frac{1}{2}mv_i^2 + \frac{1}{2}kx_i^2 = \frac{1}{2}mv_f^2 + \frac{1}{2}kx_f^2$$

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$$v^2 = gh$$

$$v = \sqrt{gh}$$

c. A conducting circular loop of radius 0.250 m is placed in the xy -plane in a uniform magnetic field of 0.360 T that points in the positive z -direction, the same direction as the normal to the plane

- Calculate the magnetic flux through the loop. [6 marks]
- Suppose the loop is rotated clockwise around the x -axis, so the normal direction now points at a 45.0° angle with respect to the z -axis. Recalculate the magnetic flux through the loop. [3 marks]
- What is the change in flux due to the rotation of the loop? [2 mark]

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$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

$$k = m v^2 + k x^2$$

END OF QUESTIONS

$$\frac{1}{2}mv^2 = \frac{1}{2}mgh^2$$

$$\frac{1}{2}mv^2 = mgh + \frac{1}{2}mv^2$$

$$\frac{1}{2}mgL^2 = \frac{1}{2}mg$$

$$\frac{1}{2}(gL^2) = \frac{1}{2}mgh + \frac{1}{2}mv^2$$

$$mgL^2 = mgh + mv^2$$

$$gL^2 - mgh = \frac{mv^2}{1m}$$

$$\frac{g}{m} \left(\frac{L^2}{h} \right) = v$$

$$V_{max} = \sqrt{\frac{2}{L} (A^2)}$$

V_{max} PE 0

$$\frac{1}{2}mv_i^2 + \frac{1}{2}kx_i^2 = \frac{1}{2}mv_f^2 + \frac{1}{2}kx_f^2$$

$$\frac{1}{2}mv_i^2 + \frac{1}{2}mgh_i = \frac{1}{2}mv_f^2 + \frac{1}{2}mgh_f$$



MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

MALAWI INSTITUTE OF TECHNOLOGY

DEPARTMENT OF APPLIED STUDIES

2019/2020 End of Second Semester Examinations

BMAT – 122: BUSINESS MATHEMATICS II

DATE: Wednesday 23rd December, 2020.

Time Allowed : 3 hours

EXAMINER: J. Mwaonaji

Instructions

1. This paper has 3 pages. Please check.
2. Answer **all** the questions.
3. Show **all** your working clearly.
4. Start each question on a **new** page.

Question	1	2	3	4	Total
Marks	25	25	25	25	100

Do not turn over this page until you are told to do so.

Question 1

(a) Let $\mathbf{a} = \langle 3, -4, 2 \rangle$, $\mathbf{b} = \langle 4, -1, -2 \rangle$ and $\mathbf{c} = \langle 3, -4, t \rangle$ be given vectors.

- Find the value of t given that vectors $\mathbf{a}$ and $\mathbf{c}$ are *equivalent* vectors. [1]
- Find the *dot product* of $\mathbf{a}$ and $\mathbf{b}$, i.e $\mathbf{a} \cdot \mathbf{b}$. [3]
- Find the magnitude of $\mathbf{a} + \mathbf{b}$ i.e $\|\mathbf{a} + \mathbf{b}\|$. [3]
- Find a *unit vector* in the direction of vector $\mathbf{a}$. [3]

(b) i. Define the term *annuity*. [4]

ii. State the difference between an *ordinary annuity* and a *due annuity*. [2]

iii. Mr. Sandifolo has a 6 months financial obligation with Phindu Investment Group amounting to MK42000. How can he fulfil this obligation in 6 months to fit into a *due annuity*? [4]

(c) Differentiate from first principles the function,

$$f(x) = 3x^2.$$

[5]

Question 2

(a) Use the method of matrices to solve the following system of linear equations below.

$$3x - 2y = 5$$

$$2x + y = 8$$

[6]

(b) If you invested a principal or present value (PV) of MK5,000 for 10 years at 9% interest rate compounded quarterly, how much money would you have, i.e the future value (FV)? [6]

(c) Consider the curve

$$y = 5 + 9x^2 - 6x^3.$$

i. Find the coordinates of the *turning point(s)* on the curve. [5]

ii. Use second derivative test to determine whether the points in 1(a) are *maximum* or *minimum* turning point(s). [4]

iii. Find the *point of inflection* on the cave. [4]

Question 3

(a) Let $A = \begin{pmatrix} 1 & 2 \\ -1 & 3 \end{pmatrix}$ and $B = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$. Evaluate the following expressions:

i. $3A - 2B$. [4]

ii. $A^2 - 2A + 3I$. [5]

iii. $\text{Trace}(A)$. [1]

Question 3 continues...

(b) Use any applicable rule of differentiation to find $\frac{dy}{dx}$ in the following functions.

i. $y = 2x^2 + e^{4x} + 3$. [3]

ii. $x^3 + y^2 = 10y^3$. [5]

(c) A curve has an equation $y = f(x)$ and it is given that $f'(x) = 3x^2 + 2x - 5$. Find $f(x)$ given that the curve passes through the point $(1, 3)$. [7]

Question 4

(a) Answer the following questions:

i. Explain what $I_{1985/1983}$ means in index numbers. [2]

ii. The following table gives details of *prices* and *quantities* sold (in pounds, £) of two particular items, UHD TV and Sound Bar (SB) in an electronic shop over two years.

Item	1984		1985	
	Price	Number sold	Price	Number sold
	p_0	q_0	p_n	q_n
UHD TV	£438	37	£462	18
SB	£322	26	£384	45

Find price and quantity *relatives* for both UHD TV and Sound Bar. [8]

(b) The demand function of company is $p = 42 - 0.001x$ and cost function is $C(x) = 30x + 1200$, where x is the number of units demanded.

i. Find the profit function $P(x)$ given that given that $P(x) = R(x) - C(x)$ and $R(x) = x.p$ [5]

ii. Find the marginal profit function $M(x)$ given that given that $M(x) = \frac{dP}{dx}$. [2]

(c) Evaluate the following limits if they exist.

i. $\lim_{x \rightarrow \infty} \left(\frac{3x^3 + 2x + 1}{5 - 2x^3} \right)$. [3]

ii. $\lim_{t \rightarrow -1} \left(\frac{t^2 + 3t + 2}{t^2 - t - 2} \right)$. [5]

END OF EXAMINATIONS.

(b) Use any applicable rule of differentiation to find $\frac{dy}{dx}$ in the following functions.

i. $y = 2x^2 + e^{4x} + 3$. [3]

ii. $x^3 + y^2 = 10y^3$. [5]

(c) A curve has an equation $y = f(x)$ and it is given that $f'(x) = 3x^2 + 2x - 5$. Find $f(x)$ given that the curve passes through the point $(1, 3)$. [7]

Question 4

(a) Answer the following questions: [2]

i. Explain what $I_{1985/1983}$ means in index numbers.

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Item	1984		1985	
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ii. $\lim_{t \rightarrow -1} \left(\frac{t^2 + 3t + 2}{t^2 - t - 2} \right)$. [5]

END OF EXAMINATIONS.



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MALAWI INSTITUTE OF TECHNOLOGY
APPLIED STUDIES DEPARTMENT

2018/2019 Second Semester Examinations

MATH-122: Calculus I

DATE: Thursday, 01 August, 2019

TIME ALLOWED: 09:00 - 12:00 (3 hours)

EXAMINERS: Dr. M. Maliyoni, Dr. P. Mwamtobe, Dr. P. Phepa, K. Magamba & D. Makupe

Instructions

1. This paper contains **THREE** questions. Please check.
2. Attempt **ALL** questions.
3. Begin each question on a **FRESH PAGE**.
4. Scientific calculators can be used where necessary.
5. Show all your working clearly and neatly.

Question	1	2	3	Total
Marks	25	45	30	100

Do not turn over this page until you are told to do so

QUESTION ONE

(a) Convert

i) 450° to radians ✓

[3 Marks]

ii) $\frac{4\pi}{5}$ to degrees ✓

[3 Marks]

(b) Evaluate the following without using a calculator (leave your answer in surd form);

i) $\sin\left(\frac{7\pi}{12}\right)$ ✓

[4 Marks]

ii) $\sec 345^\circ$

[4 Marks]

(c) Prove the following trigonometric identities;

i) $\frac{\sin 2\theta}{\cos 2\theta + 1} = \tan \theta$

[5 Marks]

ii) $\cos 3x = 4 \cos^3 x - 3 \cos x$

[6 Marks]

QUESTION TWO

(a) Evaluate $\lim_{x \rightarrow 3} \frac{x^2 - 9}{3 - x}$ ✓

[4 Marks]

(b) Determine whether or not the function given below is continuous at $x = 4$. Show your working.

$$f(x) = \begin{cases} 2x + 3, & x \leq 4 \\ 7 + \frac{16}{x}, & x > 4 \end{cases}$$

[5 Marks]

(c) Differentiate $f(x) = \sqrt{x}$ with respect to x from first principles. ✓

[6 Marks]

(d) Let $y = e^{-4x} \tan 8x$. Find $\frac{dy}{dx}$. ✓

[4 Marks]

(e) Use implicit differentiation to find y' for the function $\ln 2x + 6y^2 = x^3 \cos 4y$. ✓

[5 Marks]

(f) Let $f(x) = x^3 - 3x + 5$.

i) Find the critical points of f . ✓

[4 Marks]

ii) Identify the interval on which f is increasing and on which f is decreasing. ✓

[4 Marks]

iii) Determine the nature of the critical points found in 2f(i).

[3 Marks]

iv) Use the results in 2f(i), 2f(ii), and 2f(iii) to sketch the graph of f .

[3 Marks]

(g) A farmer has 100m of wire which he wants to fence in a rectangular field. What is the greatest area that he can enclose?

[7 Marks]

QUESTION THREE

(a) Evaluate the following indefinite integrals:

i) $\int t \sqrt{t} dt$ [4 Marks]

ii) $\int (x^{-1} + e^{2x} - 3) dx$ [4 Marks]

iii) $\int x \ln |x| dx$ [6 Marks]

(b) Evaluate each of the following definite integrals:

i) $\int_1^2 2x e^{x^2} dx$ Correct your answer to 4 decimal places [6 Marks]

ii) $\int_2^3 \frac{2x+3}{x^2-1} dx$. Correct your answer to 4 decimal places [10 Marks]

End of question paper

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MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

Malawi Institute of Technology (MIT)

Department of Applied Studies

Year One

End of Semester Two Examination

MAT 122: CALCULUS I

23rd December, 2020

Time: 9:00 - 12:00 (3 hours)

INSTRUCTIONS

- This paper contains ⁴three questions. Please Check
- Answer all questions.
- Justify your working. Unsupported answers will lead to loss of marks.
- Use of calculators is allowed.

(Do not start writing until you are told to do so)

Question One (25 marks)

(a) Evaluate the following limits

(i) $\lim_{x \rightarrow 3} \frac{2x^2 - 18}{x - 3}$ [4 Marks]

(ii) $\lim_{x \rightarrow \infty} \frac{x^3 - 4x + 3}{5x^3 - 3 - x^2}$ [6 Marks]

(iii) $\lim_{x \rightarrow 0} \frac{\sin(3x)}{x}$ [4 marks]

(iv) $\lim_{x \rightarrow 0} \left(\frac{x-4}{x} \right)^{4x}$ [5 marks]

(b) Determine whether the function given below is continuous at $x = 1$. Show your working.

$$f(x) = \begin{cases} 7x - 2 & , x \leq 1 \\ 5x^2 & , x > 1 \end{cases}$$

[6 Marks]

Question Two (21 marks)

(a) Find $\frac{dy}{dx}$ from first principles, if $y = x^2 + 2x$ [5 marks]

(b) Differentiate each of the following functions with respect to x , by using any appropriate method;

(i) $y = x^3(2x + 1)^2$ [4 Marks]

(ii) $y = \frac{(x+1)^2}{1-x^2}$ [5 Marks]

(iii) $y = (1 - 4x^5)^{14}$ [3 marks]

(iv) $y = \ln(3x^2)$ [3 marks]

(v) $y = e^{3x^2+6}$ [3 marks]

Question Three (30 marks)

(a) A farmer has 100 m of fencing. S/he wishes to construct a rectangular field to a side of a straight wall to avoid using a lot of fencing material.



$$A = L \times W$$

$$100 = L \times W$$

$$W = 2 \times 2x$$

$$L = 100 - 2x$$

$$A = (100 - 2x)x$$

$$100x - 2x^2$$

$$A = 100 - 4x$$

$$-4x$$

- (i) If x is the width of the field and y is the area of the field, express the area of the field in terms of x . [2 marks]
- (ii) Calculate the width that can be chosen for maximum area. [4 marks]
- (b) Find gradient of the tangent to curve $y^2 = 4x^2$ at the point $(1, 2)$. [6 marks]
- (c) Show that $\frac{d}{dx} \tan x = \sec^2 x$. [6 marks]
- (d) Use the results in 3(c) to find $\frac{dy}{dx}$ if $y = \tan(4x + x^2)$ [3 marks]
- (e) Find the equation of the tangent and normal to the curve $y = x^2 - 5x + 6$ at $x = 2$ [7 marks]

Question Four (24 marks)

(a) Evaluate the following integrals

(i) $\int_1^3 (x^2 + 1) dx$ [5 Marks]

(ii) $\int \frac{(x-2)^2}{x^2} dx$ [4 Marks]

(iii) $\int x e^{-x} dx$. [4 Marks]

(iv) $\int \frac{\cos x}{1 - 4 \sin x} dx$. [5 marks]

Hint: Let $u = 1 - 4 \sin x$

(b) Find the area enclosed by the graphs of $y = x$ and $y = x^2$ [6 Marks]



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MALAWI UNIVERSITY OF SCIENCE AND TECHNOLOGY

BASIC SCIENCES DEPARTMENT

END OF SEMESTER TWO EXAMINATIONS

LCOS-122: LANGUAGE AND COMMUNICATION STUDIES II

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DATE: 20th JUNE, 2016

JUNE, 2016 TIME: 09:00AM – 12:00PM (3 hours)

INTERNAL EXAMINERS: M.C. KAJOMBO, P. JIJAJIYA, & K.K. MKANDAWIRE

EXTERNAL EXAMINER: DR. P. MHAGAMA

INSTRUCTIONS

1. This paper has eleven (11) questions; please check.
2. Answer all in Section A and any other three in Section B.
3. Marks for each question are indicated against it.
4. Marks will be deducted for mistakes in spelling, grammar and punctuation.

9

Section A

Answer all questions in this section.

QUESTION 1

- a) What do you understand by the following terms? [5]
- i. Noise
 - ii. Ambience
 - iii. Feedback
 - iv. Interactants
 - v. Illustrators
- b) What role do the following play in communication process? [2]
- i. Interactants
 - ii. Environment

QUESTION 2

Briefly explain how the awareness of the following characteristics of nonverbal communication can enhance your communication competence: [6]

- a) Ambiguity
- b) Continuous
- c) Unconscious

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QUESTION 3

- a) What do you understand by the following terms in visual communication?

- i. Sensing [1]
- ii. Perceiving [1]
- b) Explain the difference between indexical and symbolic signs. [2]
- c) With examples, explain two advantages and one disadvantage of using a model over a real object during a presentation. [3]

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QUESTION 4

- a) Write a simple definition for each of the following objects:
 - i. A beaker [1]
 - ii. A microscope [1]
- b) Mention the four main steps followed when writing simple descriptions for simple objects. [4]

QUESTION 5

- a) Briefly explain what the writer should consider when writing following the PRC Triangle of Communication. [6]
- b) What is the difference between the two forms of business correspondence: a memo and a business letter? [2]

QUESTION 6

- a) Give one example of a situation where one has to read when presenting a speech. [1]
- b) Give one reason why one should read in the situation above. [2]
- c) Mention three advantages of reading as a source of one's speech? [3]

Section B

Answer any three questions from this section.

Question 7

In communication, choice of a channel is one of the major elements to consider for one to achieve effective communication. However, one has to consider a number of factors when choosing the channel. Discuss any four factors to consider when choosing a channel and explain why they are important. [20]

QUESTION 8

Discuss any four reasons why it is important to observe nonverbal cues in a communication setting? [20]

QUESTION 9

Write an essay in which you explain how your knowledge of the concepts of expectation, selectivity and habituation would help you in preparing visual aids for a presentation. [20]

QUESTION 10

Discuss the major parts of a technical report. [20]

QUESTION 11

a) Complete the following memo [15]

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MEMORANDUM

To : HoD Basic Sciences

Cc :

From :

Date :

Subject :

I am unable to attend lectures this week due to sickness.

Enc.

- b) Explain how the memo you have written in 11a above meets the conditions set out by the KISS Principle. [5]

End of Question Paper

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2



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DEPARTMENT OF BASIC SCIENCES

SECOND SEMESTER EXAMINATION

HAPH-121: HUMAN ANATOMY AND PHYSIOLOGY I

100 MARKS

Date: Friday, 22nd June, 2016

Time: 9.00-12.00 noon (3 hours)

EXAMINERS: G. Moyo, P. Makocho and P. Samson

Instructions:

1. Check that this paper has got 3 printed pages and 15 questions. Please check!
2. Answer ALL questions in section A and any four in section B the using the answer book provided.

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SECTION A: Answer all questions

- 1 (a) Briefly describe the process of simple filtration in the formation of urine. [5]
- (b) Explain the following giving an example for each: [4]
- i. High-threshold substances
 - ii. Non-threshold substances
- 2 (a) The effects of an alcoholic drink are felt soon after drinking and rapid pain relief can be gained from taking aspirin. Explain why these substances have such a rapid effect when ingested. [1]
- (b) Explain the role of the following in the stomach: [4]
- i. The cardiac sphincter.
 - ii. Internal stomach folds (rugae).
 - iii. Secreted mucus
- (c) Explain how the villi and microvilli assist in the efficient absorption of nutrients.
- 3 Describe the three systems of blood circulation. [6]
- 4 (a) Explain the functional relationship between the peripheral nervous system and central nervous system. [3]
- (b) State any two differences between thalamus and hypothalamus. [4]
- 5 (a) What is the difference between expiration and inspiration? [2]
- (b) Mention any two accessory muscles of respiratory system and explain why it is important at other times to use them? [3]
- (c) Explain how inspiration qualifies as an active process unlike expiration. [4]
- 6 (a) Explain what would happen if the ovaries were surgically removed from a girl before she reached puberty stage (give two points). [2]
- (b) State any two differences between spermatogenesis and oogenesis. [2]

- 7 Describe the steps of fracture repair. [8]
- 8 How different is the cell structure of a skeletal muscle cell from any other cells in the body? [5]
- 9 For humans sleep is a vital indicator of overall health and well-being. Stimulants like coffee and energy drinks, alarm clocks, and external lights—including those from electronic devices—interferes with our “circadian rhythm” or natural sleep/wake cycle. Explain the control mechanism of sleep /wake cycle in the body. [7]

SECTION B: Answer any four questions

- 10 Briefly describe five factors that affect pulse. [10]
- 11 What are the possible metabolic process of the end product of carbohydrate digestion (glucose)? [10]
- 12 Explain the accessory structures of eye with their importance. [10]
- 13 The sense of smell and taste are conducted to the nervous system through the process of signalling within the cell. How is transduction of an odorants to the receptors different from the transduction of taste? [10]
- 14 Describe how respiration is controlled by chemicals and nervous system. [10]
- 15 Contrast the anatomy and physiology of the parasympathetic and sympathetic divisions. [10]



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ACADEMY OF MEDICAL SCIENCES

DEPARTMENT OF BIOLOGICAL SCIENCES

END OF SEMESTER 2 EXAMINATION

HAPH-121: HUMAN ANATOMY AND PHYSIOLOGY

100 MARKS

Date: 15th August, 2018

Time: 09h00- 12h00 (3 hours)

EXAMINERS: T. Msiska, I. Kondowe-Chinangwa and M. Nyando

Instructions:

1. Check that this paper has got five printed pages and eight questions
2. Answer ALL questions using the answer book provided.

Answer ALL questions

Nervous system

1.

a) Name the nerves that are involved in the following

i. flexion of the thigh

[1 mark]

ii. sexual stimulation and responses

[1 mark]

b) Which structures make up the following?

i. Brain stem

ii. Diencephalon

[4 marks]

c) Name the cranial nerve that:

i. Innervate one of the six muscles responsible for moving the eye

[1 mark]

ii. Supplies motor innervation to muscles of mastication

[1 mark]

000TTÁFAVÓGÓVÁH

d) Mention the three structures that help to protect the brain

[3 marks]

e) What name is given to the cluster of cells that are involved in the production of cerebral spinal fluid?

[1 mark]

f) Injury of which nerve could cause paralysis of the serratus anterior muscle?

[2 marks]

g) Why is it that some nerves are called mixed nerves?

[2 marks]

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Cardiovascular System

2.

- a) Mention two blood vessels the heart uses to achieve venous return. [2 marks]
- b) What are the three tributaries of the coronary sinus? [3 marks]
- c) What is the importance of the azygous system relative to the inferior vena cava? [2 marks]
- d) Describe the structure of a blood capillary? [2 marks]
- e)
 - i. What is the primary function of capillaries? [1 mark]
 - ii. Why is the structure of the capillary important? [2 marks]
- f) Identify the three major veins that drain blood from the head [3 marks]

Muscular System

3.

- a) List the muscles forming the rotator cuff. [4 marks]
- b) List the muscles that cause flexion and extension of the elbow. [5 marks]
- c) What muscles produce supination and pronation of the forearm? [2 marks]
- d) Mention the muscles that make the hamstring. [3 marks]

Skeletal System

4.

- a) Name the four major sutures of the skull and the bones they connect. [8 marks]
- b) What are the functions of the vertebral column? [4 marks]
- c) Describe scoliosis, kyphosis, and lordosis. [3 marks]

36
8
44

Endocrine System

5. Why are the ovaries and testes classified as endocrine glands as well as reproductive organs? [10 marks]

Produce estrogen Leydig cells
Produce ~~testosterone~~ Oestrogen Progesterone Female
the produce sperm
6. they store ovum

Digestive System.

- a) What are the structural features that increase the luminal surface of the small intestine? [3 marks]
- b) In a clinical setting how can you use the McBurney's point to locate the appendix? [3 marks]
- c) Give the function of the following types of cells in the small intestine

- i. Goblet cells ✓ [1 mark]
- ii. Enterocytes ✓ [1 mark]
- iii. Enteroendocrine cells ✓ [1 mark]
- iv. Paneth cells ✓ [1 mark]

Reproductive System.

7.

- a) What is the function of the following
- i. Epididymis ✓ [1 mark]
 - ii. Seminiferous tubules ✓ [1 mark]
 - iii. Spongy spongiosum ✓ [1 mark]
- b) Describe, 'perineum.' ✗ [2 marks]
- c) What is the clinical significance of the perineum in mothers? ✗ [2 marks]
- d) Give the function of the following accessory structures
- i. Prostate gland (2) [1 mark]
 - ii. Seminal vesicle. [1 mark]
 - iii. Bulbourethral gland (cowper's gland). [1 mark]

Respiratory system

8.

- a) Mention four muscles that are involved in respiration (4) [4 marks]
- b) What are the basic differences between cortical and juxtamedullary nephrons? ✗ [2 marks]
- c) Describe blood supply to and from the lungs. (3) [4 marks]

[END OF PAPER]



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END OF YEAR EXAMINATION

Human Anatomy and Physiology

HAPH-121

100 MARKS

Date: 25th July, 2019

Time: 13:30- 16:30 (3 hours)

EXAMINER: L. Afune, D. Nkhota, T. Msiska & Nyando, PhD

Instructions:

1. Check that this paper has got 4 printed pages and 4 questions
2. Answer ALL questions using the answer book provided.

Answer ALL questions

In reproduction

1a) Define

- i. Ovarian cycle [2]
 - ii. Follicular phase [2]
- b) i. Name the final phase of the ovarian cycle. [1]
- ii. What is the corresponding phase for the answer given in part b)i in the uterine cycle. [1]
- c) Give two major differences between the Graafian follicle and corpus luteum in the female reproductive system [4]
- d) Mention two pouches of peritoneum that surround the uterus [2]
- e) Describe the anatomical location of the perineum [2]
- g) Name the arteries that supply blood to the stratum functionalis layer of the uterus [1]

Regarding the nervous system

- 2a) Name the two principal cell types of the nervous system [2]
- b) Give the functions of the following cells:
- i. Astrocytes (two functions) [2]
 - ii. Microglia (one function) [1]
 - iii. Ependymal (two functions) [2]
- c) Which nerve innervates the muscles that adduct the thigh? [1]
- d) Which nerve is involved in the flexion of the thigh [1]

- e) Which nerve plays an important role in sexual stimulation and responses [1]
- f) Which nerve innervates the gastrocnemius [1]
- g) Name the three branches of the trigeminal nerve. [3]
- h) Name the cranial nerve that:
 - i. Innervates one of the six muscles responsible for moving the eye. [1]
 - ii. Supplies motor innervation to muscles of mastication [1]
- i) Mention the three structures that help to protect the brain [3]

Concerning the endocrine System:

3a) Differentiate between the nervous and endocrine systems with regard to the control of homeostasis? [4]

- b) Distinguish between exocrine and endocrine glands [2]
- c) List three organs or tissues that are not exclusively classified as endocrine glands but contain cells that secrete hormones. [3]
- d) Mention the chemical classes of hormones, and give an example of each. [4]
- e) How are hormones transported in the blood? [4]

Regarding the digestive System:

4. Give five major peritoneal folds and the significance of each one of them [10]

In the respiratory System:

- 5a) Distinguish the parietal pleura from the visceral pleura. [2]
- b) Identify the parts of the respiratory passageway and its associated structure as labelled in Figure 1 (6 marks)

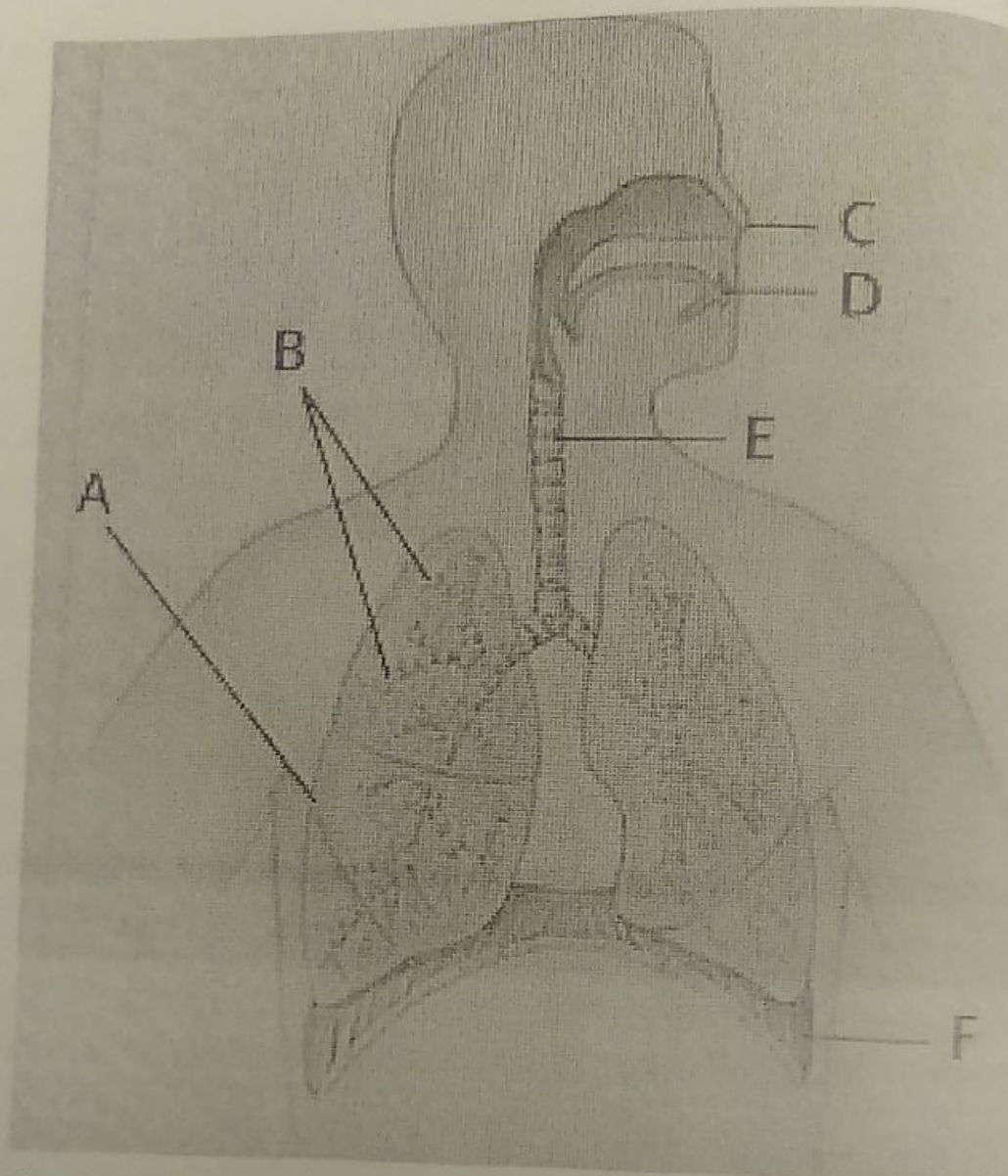


Figure 1: The respiratory system

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c) Explain the difference between external and internal respiration [4]

In the urinary system:

6. Figure 2 represents a structure of an organ in the human body, study it and answer questions that follow.

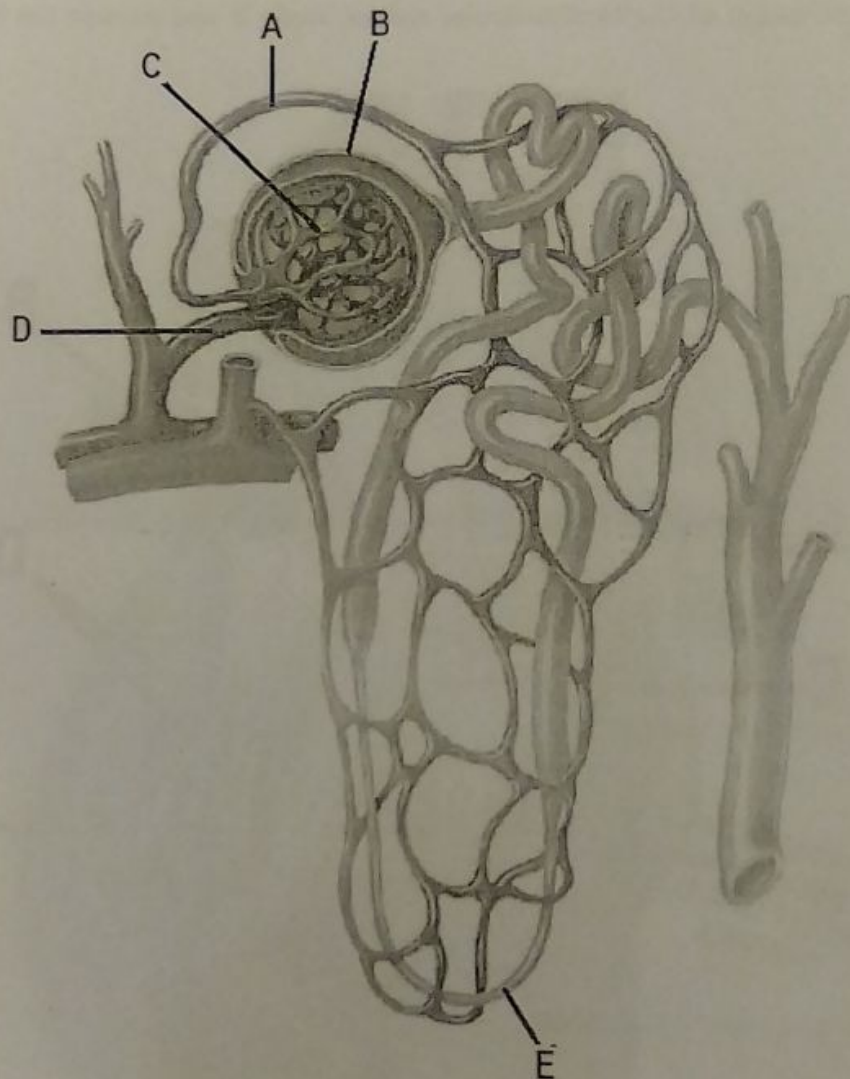


Figure 2: organ structure in the body

a) Mention the parts labeled A, B, C, D and E [5]

b) Give two differences between the parts labeled A and D [2]

c) Describe the three processes necessary for urine formation [6]

Regarding cardiovascular system:

7. Figure 3 is an organ of the cardiovascular system, study it and answer the following questions.

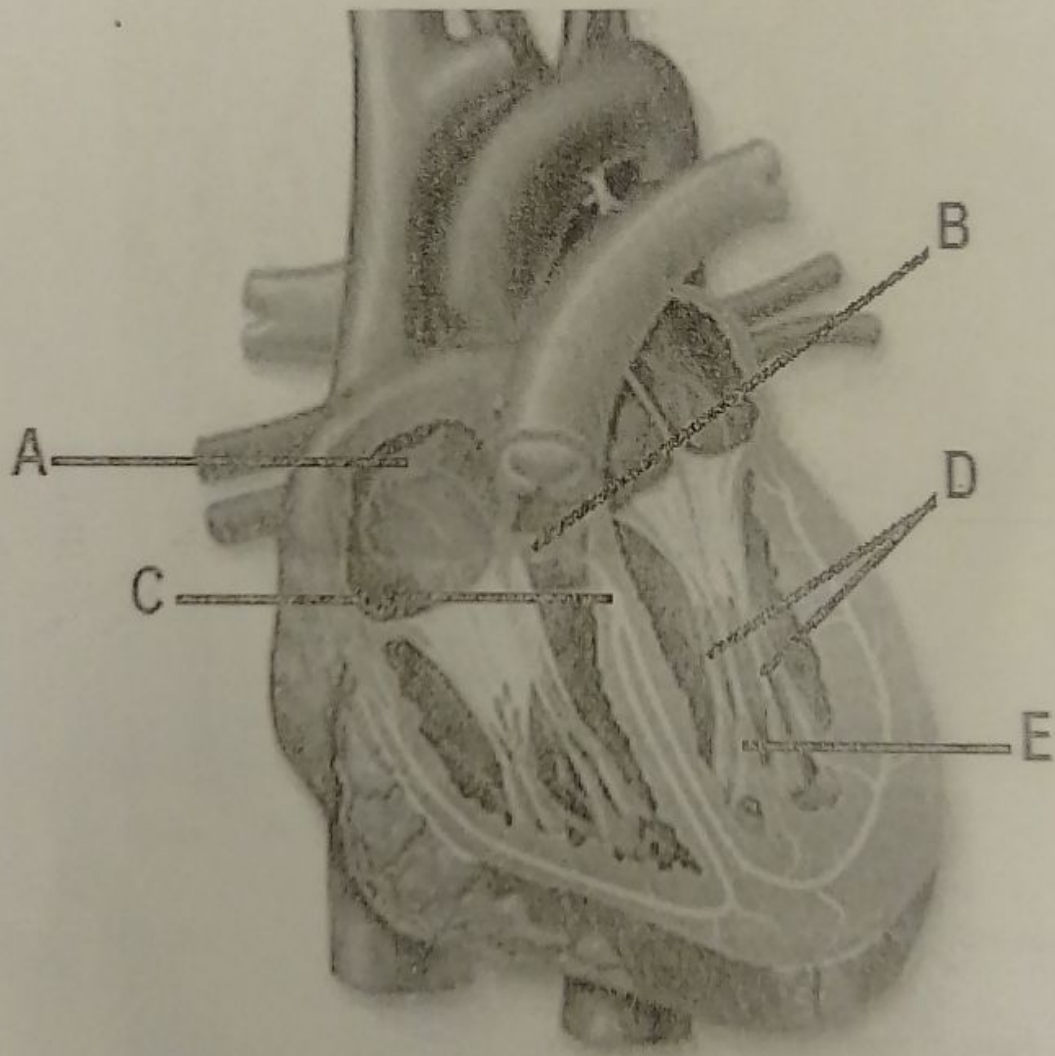


Figure 3: An organ in the body

a) Name the parts labeled A, B, C, D and E [5]

- b) Mention the main function of part labeled D [1]
- c) Explain how parts labeled A, B and C control the heart contractions [6]

[END OF PAPER]

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END OF SEMESTER 2 EXAMINATION
HAPH-121: HUMAN ANATOMY AND PHYSIOLOGY 1

100 MARKS

Date: Wednesday, 16th December, 2020

Time: 13:30- 16:30 (3:00 hours)

EXAMINERS: Dr T. Mwenyenkulu

Instructions:

1. Check that this paper has got 4 printed pages and 4 questions
2. Answer ALL questions using the answer book provided.
3. Write your registration number on the answer book provided.
4. Do not write anything on the question paper

1. Regarding the nervous system; [2 marks]
- a. Give two (2) examples of inhibitory neurotransmitters. [2 marks]
 - b. State the specific functions of the following in the human body;
 - i. Ruffini's end organ [1 mark]
 - ii. Krause end bulbs [1 mark]
 - iii. Meissner's corpuscle [1 mark]
 - iv. Merkel's disks [1 mark]
 - v. Pacinian corpuscle [1 mark]
 - vi. Free nerve endings [1 mark]
 - c. List any two (2) types of sensory impulses that are carried by the spinothalamic tract (pathway). [2 marks]
 - d. Differentiate between the dorsal column medial lemniscal system and the spinothalamic pathway. [4 marks]
 - e. What are the roles of the following in a neuron?
 - i. Dynein [2 marks]
 - ii. Kinesin [2 marks]
 - f. Explain the differences in functionality between astrocytes and oligodendrocytes. [4 marks]
 - g. Discuss how the transmission of an impulse at the synaptic junction occurs (include all the electrolytes that are relevant). [5 marks]
 - h. What is the significance of the thalamus in sensory pathways? [2 marks]
 - i. Which Brodmann areas of the brain are collectively referred to as the somatosensory cortex? [3 marks]
 - j. Explain any five (5) possible conditions/manifestations that a person might experience when there is a damage to the somatosensory cortex. [5 marks]
 - k. Explain the fate of a neurotransmitter at the synaptic junction. [4 marks]
 - l. With the aid of a diagram discuss how an action potential is generated at the post synaptic membrane. [5 marks]

2. About the digestive/gastrointestinal system

- a. Differentiate between parietal peritoneum and visceral peritoneum [2 marks]
- b. List any four parts or organs of the digestive tract that are located retroperitoneally [4 marks]
- c. List the three salivary glands that are present in a human being. [3 marks]
- d. Which component of the motor nervous system controls the activities of salivary glands? [2 marks]
- e. List any two (2) functions of bile salts. [2 marks]
- f. Discuss the defecation reflex. [4 marks]
- g. Discuss the role of enterogastrones in relation to high fat diet digestion. [4 marks]
- h. Give four (4) factors that regulate the motility of the contents of the stomach. [4 marks]
- i. What are the four (4) functions of the hydrochloric acid that is present in the gastric juice? [4 marks]

3. About the cardiovascular system

- a. Discuss the term 'functional syncytium' with regards to cardiac muscle contraction. [4 marks]

- b. What causes heart sounds? [1 mark]

4. Concerning general knowledge application

- a. Discuss the interdependence that exists between the gastrointestinal system and the cardiovascular system. [10 marks]

- b. Compare and contrast the nervous system and the endocrine system [10 marks]

[END OF QUESTION PAPER]